# The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy

Master's thesis, reading copy. Chapters 6 and 7 are drafts, and Section 6.5 waits for results due from 27 October 2026.

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# Abstract

**The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy**

Figures about the digital economy stand for activity that no one observes directly. A platform's revenue stands for the commerce it carries, and a survey's count of sellers stands for the businesses selling online. This thesis asks how much activity remains invisible when platforms are measured through their revenue, and when that choice changes the conclusion about growth. It studies Indonesia, using platform reports, BPS's survey of online sellers and national figures.

Revenue shows only a small part of the commerce. In 2023, three major platforms in Indonesia kept about US$3 billion of the US$43 billion that passed through them. Revenue also stopped tracking the commerce when platforms changed the share of each sale they kept. Between 2022 and 2024, Indonesian and regional platforms changed that share about three times as much as 27 platforms abroad. So the hypothesis that revenue grows in proportion to the commerce is rejected.

The national count of online sellers rose by 27 percent in 2023. A test registered before the survey files were opened shows that first-time entry explains only about half of that rise. More businesses account for most of the growth in online sales on the published figures. If the rest of the rise is read as wider survey coverage, that conclusion sits on its threshold.

The two results describe one problem. A figure stops tracking the activity it stands for when the link between them changes, through a platform's take rate or a survey's reach. The thesis also measures how far Indonesia's recent platform rules can reach: the new marketplace tax can collect from at most about 1 in 20 online sellers.

**Keywords:** digital platforms; take rate; e-commerce measurement; transaction value; online sellers; Indonesia

*[Chinese abstract (中文摘要), Chinese keywords and the Chinese title page are required by YZU and are left for the researcher. The proposal's Chinese title was 隱形帳簿：量化印尼平台經濟中的隱形缺口.]*

# Symbols and Abbreviations

Each symbol is defined in Chapter 3, where it first appears in a formula. The subscripts i and t mark the platform and the year.

| Symbol | Meaning | First defined |
|---|---|---|
| V | Transaction value: everything bought through a platform in a period | Table 3.1 |
| R | Revenue: what the platform records as its own income | Table 3.1 |
| m | Take rate, R divided by V: the share of each sale the platform keeps | Table 3.1 |
| f | Fee rate: revenue before customer incentives, divided by V | Table 3.1 |
| s | Incentive rate: customer incentives divided by V | Table 3.1 |
| W | Invisible wedge, V minus R: what a revenue figure leaves out | Equation (1) |
| E | Ecosystem ratio, W divided by R, equal to 1/m − 1 (the author's own measure) | Equation (2) |
| g(·) | Growth from one year to the next, as a share of the first year | Table 3.1 |
| D | Growth divergence, g(R) minus g(V) | Equation (3) |
| Δln X | Change in the natural logarithm of X from one year to the next | Equation (5) |
| N | BPS's count of businesses selling online | Equation (7) |
| Y | BPS's total online sales value | Equation (7) |
| MV | A listed platform's market value | Equation (8) |
| $\beta_V$, $\beta_m$ | How market value moves with transaction value and with the take rate | Equation (8) |
| H1, H2 | The two hypotheses | Section 3.4 |

The text spells out most names. These abbreviations remain.

| Abbreviation | Meaning |
|---|---|
| BPS | Badan Pusat Statistik, Statistics Indonesia |
| GMV, GTV, TPV | Gross merchandise value, gross transaction value, total payment volume: company names for transaction value |
| IASB, IFRS 15 | International Accounting Standards Board; International Financial Reporting Standard 15, its revenue standard |
| LSEG | London Stock Exchange Group, the source of market values |
| OECD | Organisation for Economic Co-operation and Development |
| QRIS | Quick Response Code Indonesian Standard, the national QR-code payment system |
| UNCTAD | United Nations Conference on Trade and Development |
| Rp, US$ | Indonesian rupiah, US dollar |

# Chapter 1. Introduction

Digital commerce has changed how goods and services are bought and sold in Indonesia. Sellers in small towns reach buyers across the country, businesses run from mobile phones, and customers pay by QR code. The figures that measure this activity disagree on both its size and its growth.

In 2023, traders at Tanah Abang, a major wholesale market in Jakarta, said their customers had moved to sellers on TikTok Shop (AFP 2023). The government then banned transactions on social media platforms, and TikTok Shop closed in Indonesia soon after (Ministry of Trade of the Republic of Indonesia 2023; Mandagie 2023). That year the national statistics office, BPS, reported that online sales rose by about 41 percent (BPS 2025a). The central bank's measure of online shopping fell by about 5 percent (Bank Indonesia 2023; KabarBursa 2024). The two official figures moved in opposite directions.

Tokopedia, an Indonesian online marketplace, shows revenue and commerce moving apart inside one company. Between 2022 and 2023, its net revenue rose by 53 percent while the value of transactions on the platform, its transaction value, fell by 9 percent (GoTo Gojek Tokopedia 2024a, 2024b). The two are different amounts. A platform keeps only part of each sale, as commissions and fees. The rest is the seller's receipts, the driver's fare or another amount that passes through. This thesis calls the difference the invisible wedge. The term means what a revenue-only measure leaves out. It does not mean activity that goes unrecorded.

The share a platform keeps is not fixed. When it rises, revenue can grow even while transaction value falls. In Tokopedia's accounts, about 60 percent of the extra revenue came from smaller discounts and incentives, which a platform pays for out of what it earns (GoTo Gojek Tokopedia 2024b).

Platform and online sales figures carry weight well beyond Indonesia. Businesses in 43 economies sold almost US$27 trillion online in 2022, by the United Nations' estimate (UNCTAD 2024), and platforms carry a growing part of that trade. Investors value the companies that run them largely on reported growth. If part of that growth is a platform keeping more of each sale, then part of the share price rests on a pricing decision. Governments use the same figures to size their digital economies and to tax online sellers.

Research explains why platform revenue and national sales figures can differ, but not when a figure stops moving with the activity it represents, or by how much. Answering that takes two things that are rarely found together. One is a platform's revenue and transaction value over several years. The other is a national count of online sellers whose underlying answers can be examined.

In Indonesia, several platforms report revenue and transaction value each year, and BPS asks online sellers when they began and how their sales changed. The stakes are concrete. For 2024, BPS gives about US$81 billion for online sales and the central bank about US$31 billion, more than two to one (BPS 2025b; Bank Indonesia 2025). The government has set new rules on platform commissions, discounts and tax. Its tax on marketplace sellers can raise about US$60 million a year on one measure of the market and about US$300 million on another, before exemptions (Ministry of Finance of the Republic of Indonesia 2025; author's calculation). Most online sellers work through messaging and social media, outside the marketplaces, so platform and survey figures are likely to part. The events of 2023 suggest that they did.

## 1.1 Research question and objectives

This thesis sets out to change how the digital economy is measured. Most online commerce is missing from the platform figures on which companies are valued and sellers are taxed, and millions of the people who sell online are outside those figures altogether. Decisions worth trillions of dollars rest on

numbers that show a fraction of the activity they are taken to describe. The ambition of this thesis is to bring the rest into view: to show how much of the online economy the current figures miss, where it is, and who is in it.

Measuring what the current figures miss begins with one question: how much economic activity remains invisible when digital platforms are measured through their reported revenue, and when does that choice change the conclusion about growth? The question has two halves. The first concerns size, which is how much of the activity lies outside the figure. The second concerns growth, which is whether the figure still moves with the activity once prices or coverage change. A large wedge would matter less if it stayed in proportion; the second half asks whether it did.

The research question is addressed through three objectives. The first is to measure the invisible wedge across the platform histories available from 2020 to 2025. The second is to reconcile selected large movements with disclosed changes in monetization, customer incentives, revenue recognition and reporting scope. The third is to compare the platform evidence with national e-commerce and payment figures, to test whether they give the same account of economic scale and growth.

The thesis tests two hypotheses. H1 says that a platform's revenue grows in proportion to its transaction value. H2 says that national e-commerce growth comes mainly from more businesses selling online, and less from higher sales per business. An entry test then asks whether a rise in the count of online sellers comes from sellers entering for the first time. A further analysis estimates how many sellers a rule built on platform records can reach.

In brief, revenue shows only a small part of the commerce that platforms carry. In BPS's survey, the marketplaces carry only about a fifth of online sales value. In Indonesia, platform revenue also grew faster than the commerce behind it, mainly because platforms kept a larger share of each sale and not because buyers bought more. National growth came mainly from more sellers, as H2 states, but about half of the 2023 rise in the count of sellers is not explained by first-time entry.

## 1.2 Contributions

The thesis makes three contributions, one for each group that relies on these figures: investors, statistics offices, and the authorities that tax and regulate online selling.

For those who value platforms, the thesis shows how small a part of the commerce a platform's revenue records. This matters because market values of internet firms track revenue (Bowen, Davis and Rajgopal 2002). It matters for growth as well as scale: the larger the wedge is next to revenue, the more a small change in fees or discounts moves revenue growth. When the take rate moves, market value per dollar of revenue and market value per dollar of commerce change by different amounts. The thesis also traces why revenue moved, which gives investors a check on whether reported growth came from more commerce or from a larger share of each sale. A prediction registered in advance then tests how market value moves with each of the two.

For those who measure the economy, the thesis shows where the commerce outside the platforms sits and tests what a rise in the count of online sellers represents. Most online sellers are small businesses that keep no financial statements. Each reports the year it began selling online, so the survey shows how much of a rise is first-time entry. The test was specified before the survey files were opened.

For those who design tax and regulation, the thesis measures how far rules based on platform records reach. A rule built on platform records reaches only the sellers who use platforms, so where most sellers work outside them, such rules cover a minority. The thesis also sets each recent rule, from the cap on ride-hailing commissions to the limit on discounts, against the figure it acts on. Several other countries have adopted similar rules, so the same question applies there.

Earlier work has already compared platform transaction values with national e-commerce statistics. The contribution of this thesis is connected evidence for one country. It draws on the platforms' accounts, the sellers' own survey responses and the rules that act on those figures. Across all three, one point recurs. A figure can stop tracking the activity it stands for when the link between them changes, through a platform's take rate or a survey's reach.

## 1.3 Structure of the thesis

The thesis moves from ideas to evidence and then to what follows from it. Chapter 2 reviews why platform revenue differs from the commerce behind it, how online selling is measured, and what platform records can show. It ends with the questions the literature leaves open. Chapter 3 turns those questions into measures, two hypotheses, the entry test and four further analyses. Chapter 4 describes the data, which come from platform reports, BPS survey files and national and market figures, and the tests that use them. Chapter 5 gives the results, first for the platforms and then for the national figures. Chapter 6 asks how far Indonesia's recent rules reach and what the findings mean for other countries. Chapter 7 concludes and states the limits of the evidence.

# Chapter 2. Literature Review

Digital platforms sit between economic activity and the figures used to measure it. A platform carries a sale between a buyer and a seller. It records part of the payment as its own revenue, and it keeps a record of the whole sale. Different bodies of research study these steps. Platform economics and accounting explain why revenue is only part of each payment (Section 2.1). Official statistics explain how online sales across the country are measured (Section 2.2). Research on third-party information explains what a platform's records can show to tax authorities (Section 2.3). Each body of research explains why the figures differ, but none measures how far they differ in practice. Section 2.4 sets out this research gap.

## 2.1 Platform economics and revenue recognition

A platform reports two figures that describe its business: the value of the commerce it carries, and the revenue it keeps from them. Investors price that revenue: market values of internet firms track it, and share prices respond to revenue news beyond earnings news (Bowen, Davis and Rajgopal 2002; Davis 2002). Revenue is a fair guide to the commerce only if it keeps a steady share of that commerce.

Platform economics explains why revenue is only a small share of the commerce. A platform serves two groups that need each other, such as shoppers and merchants, or riders and drivers. It sets its charges for both groups together and earns fees for arranging trades, not the value of each sale (Rochet and Tirole 2003; Armstrong 2006; Evans and Schmalensee 2016). The level of its charges and the split between the two groups are choices, so the platform can take more or less revenue from the same commerce (Rochet and Tirole 2003; Caillaud and Jullien 2003).

Accounting rules add a second reason. Under the international standard, International Financial Reporting Standard 15 (IFRS 15), a company that only arranges a sale records its fee and not the full payment. If a rider pays 100 for a trip and the driver keeps 80, such a company records 20. Discounts and vouchers that it pays for are normally deducted from its revenue (International Accounting Standards Board 2014). So revenue can rise when a platform raises its fees or cuts its discounts, with no change in what customers buy.

Platform economics and accounting explain why the share is small and why it can move, but they leave its actual size and movement to be measured. The pricing models were built to explain how a platform sets its charges, so they predict the direction of a change, not its size in a given market. The accounting studies asked how internet firms around 2000 chose to report revenue, not how far that revenue drifted from the commerce behind it (Bowen, Davis and Rajgopal 2002). Studies of how investors valued those firms, and how analysts forecast their revenue, relied on web traffic as their measure of activity (Trueman, Wong and Zhang 2000, 2001; Rajgopal, Venkatachalam and Kotha 2003). Transaction value also needs care, because each company defines it for itself. A figure of this kind loses information when its definition changes (Givoly, Li, Lourie and Nekrasov 2019). Testing whether revenue stays a fair guide needs the reported revenue and transaction value of the same platforms, over several years and on one definition. The studies reviewed here do not make that comparison, and none covers Indonesian platforms.

## 2.2 Measuring the digital economy

A country's online commerce is larger than what platforms carry, because many sellers trade outside platforms. Several producers measure it, and each counts something different. Industry reports add up the value of goods sold through platforms (Google, Temasek and Bain 2024). Statistics offices, such as

BPS in Indonesia, survey businesses about their online sales (BPS 2025a, 2025b). Central banks count electronic payments (Bank Indonesia 2025). The resulting values change with the definition used (UNCTAD 2023, 2024).

Official statistics address the question in two ways. National accounts separate the sale between a buyer and a seller from the platform's fee, and they count only the fee as the platform's output (Ahmad and Schreyer 2016; International Monetary Fund 2018; Organisation for Economic Co-operation and Development [OECD] 2023; United Nations et al. 2025). Research on measurement also offers a method: test a figure against the activity it stands for, as studies of night-time lights do for national income (Henderson, Storeygard and Weil 2012). Growth in online sales can be split in the same spirit, into more sales by existing sellers and more sellers entering. Trade economists split a country's exports in this way, into more of each good and more kinds of goods (Hummels and Klenow 2005).

Official statistics and measurement research say what to count and how to check one figure against another. Neither follows platform figures and national figures through time in one country. National accounting guidance defines the platform's output, and it does not track how platform commerce and national online sales grow. The test of one figure against another has been applied to national income and night-time lights. Among the studies reviewed here, it has not been applied to platform commerce and national online sales.

The growth split has its own difficulty, because it rests on a count of sellers. Such counts can mislead for known reasons. A change in survey design can create a break that looks like real change (van den Brakel, Smith and Compton 2008; van den Brakel, Zhang and Tam 2020). People tend to report past events as more recent than they were (Neter and Waksberg 1964). Small firms that stop are more often young firms with young owners (McKenzie and Paffhausen 2019). Published totals cannot show whether a rise in the count reflects new sellers or a wider survey, because they do not record when each seller began. That question needs the answers of the sellers themselves.

## 2.3 Digital records and third-party information

A platform records every sale it carries: who sold what, to whom, for how much and when. Governments have begun to use these records for tax. The OECD, a group of mostly high-income countries, set out model rules under which platforms report what their sellers earn (OECD 2020). The European Union applies these rules through its DAC7 directive (European Union 2021). Indonesia goes further and requires marketplaces to collect tax from their sellers (Ministry of Finance of the Republic of Indonesia 2025).

Tax research explains why such records matter. Reports from third parties improve compliance. In Denmark, evasion was close to zero on income that employers and banks reported, and substantial on income that taxpayers reported themselves (Kleven et al. 2011). In Chile, the paper trail of the value-added tax deters evasion, and enforcement spreads up the supply chain (Pomeranz 2015). In Brazil, rewards for asking for receipts raised the sales that firms reported (Naritomi 2019). Digital records can also hold information that conventional records lack (Berg, Burg, Gombovic and Puri 2020).

Tax research shows that platform records are valuable where they exist. Its evidence comes from settings where a third party already sees the transaction, such as an employer paying wages or a firm buying from another firm. Online selling differs, because many sellers trade through messaging apps and social media, where no marketplace records the sale (BPS 2025a). How far a rule built on platform records can reach depends on how much selling happens on platforms. The studies reviewed here do not estimate that share for Indonesia.

## 2.4 Research gap

The shortfalls identified in Sections 2.1 to 2.3 define the research gap. Table 2.1 sets them side by side.

**Table 2.1. What the literature establishes and where it stops**

| Section | What it establishes | Main sources | Where it stops |
|---|---|---|---|
| 2.1 Platform economics and accounting | Revenue is a fee that is a small share of the commerce, and the share can move with pricing, discounts and accounting | Rochet and Tirole 2003; Armstrong 2006; IASB 2014 | Whether reported revenue stays a fair guide to reported transaction value has not been tested, and not for Indonesian platforms |
| 2.2 Official statistics and surveys | Platform commerce and platform revenue differ; sources differ; a count of sellers can break | Ahmad and Schreyer 2016; Henderson et al. 2012; van den Brakel et al. 2008 | Platform and national figures have not been followed through time in one country, and published totals cannot show what a rise in the count of sellers contains |
| 2.3 Third-party information | Platform records show sales, and third-party reports improve compliance | Kleven et al. 2011; OECD 2020 | How much of online selling platform records cover has not been estimated for Indonesia |

Each shortfall becomes one question for the thesis, and Chapter 3 sets out the tools that answer each.

1. **How large is the difference between transaction value and revenue, and does it last?** This follows from Section 2.1. It explains why the share is small and can move, but it does not test whether revenue stays a fair guide to the commerce.
2. **Does the choice of figure change the conclusion about growth?** This follows from Section 2.2. It shows that figures differ, but it does not test whether they grow for the same reasons, or what a rise in the count of sellers contains.
3. **How far can rules built on platform records reach?** This follows from Section 2.3. It shows that platform records are valuable, but not how much of online selling they cover.

# Chapter 3. Framework and Hypotheses

Platform activity can be observed through several figures, and these figures do not measure the same economic object. Transaction value measures the commerce a platform carries. Revenue measures the part of that commerce the platform keeps. National statistics measure online sales across all sellers, whether or not they use a platform.

The framework tests each figure against the activity it is used to represent, as studies of night-time lights do for national income (Henderson, Storeygard and Weil 2012). It starts from the difference between the value a platform processes and the revenue it recognizes. It then asks whether that relationship stays stable over time, and whether figures from other sources give the same account of online activity. Figure 3.1 shows the structure.

Figure 3.1 places the three quantities inside one another. Revenue sits inside platform commerce, because a platform keeps only part of each sale. Platform commerce sits inside all online commerce, because many sellers trade outside platforms. The first hypothesis, H1, tests the inner link between revenue and platform commerce. The second, H2, tests the outer link between platform commerce and national online commerce.

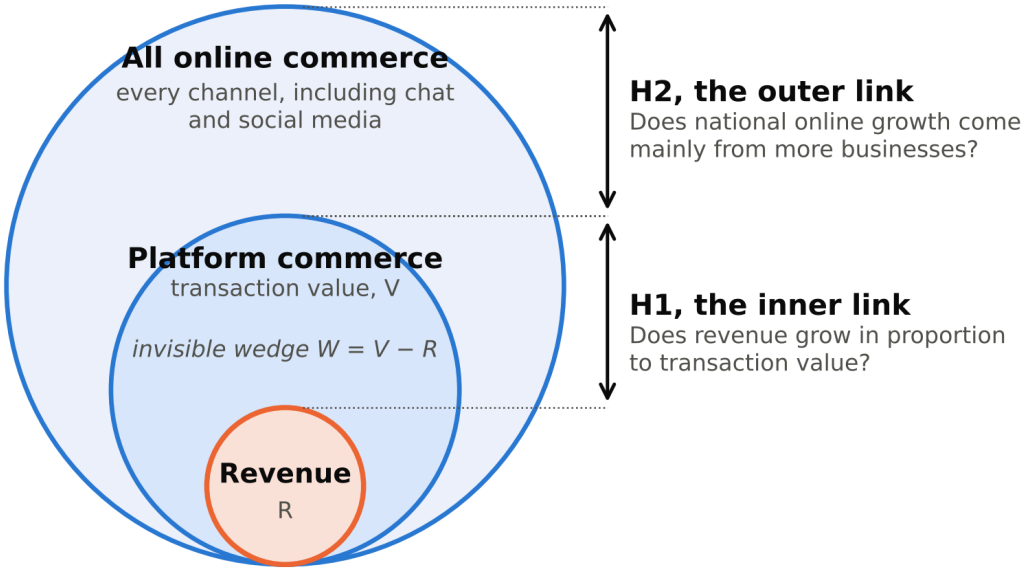



**Figure 3.1. Three nested quantities and the two links.** Revenue sits inside platform commerce, which sits inside all online commerce. H1 tests the inner link and H2 the outer link.

*Note:* The figure is illustrative, and the sizes are not to scale. The platform circle differs from the marketplace channel in BPS's survey of online sellers.

The research question asks how much activity remains invisible when platforms are measured through revenue, and when that choice changes the conclusion about growth. The literature leaves three parts of it open (Section 2.4). The framework answers each with its own tools:

1. **How large the difference between transaction value and revenue is, and whether it lasts:** three measures, a split of revenue growth, and H1 (Sections 3.2 to 3.4).
2. **Whether the choice of figure changes the conclusion about growth:** H2, the entry test and a comparison of national figures (Sections 3.4 to 3.6).

3. **How far rules built on platform records reach:** how online selling divides between marketplaces and other channels (Section 3.6).

The three objectives set in Chapter 1 sit inside this map. Measuring the wedge and tracing its large moves serve the first question, and comparing platform figures with national and payment figures serves the second. The third question is answered by a further analysis.

## 3.1 Terms and rules for comparison

Table 3.1 defines the terms used throughout the thesis, each once, with the symbol used in the formulas.

**Table 3.1. Terms and symbols**

| Term | Symbol | Meaning |
|---|---|---|
| Transaction value | V | Everything bought through a platform in a period, also called the commerce the platform carries. Companies call it gross merchandise value (GMV), gross transaction value (GTV) or total payment volume (TPV); the thesis treats these as one measure and notes where a company's scope differs. |
| Revenue | R | What the platform recognizes as its own income in its accounts. |
| Take rate | m | Revenue divided by transaction value: the share of each sale that the platform keeps. |
| Customer incentives | | Discounts, vouchers and other promotions that a platform pays for. Accounting rules normally deduct them from revenue, and Tokopedia reports the deduction as a separate line. |
| Fee rate | f | The platform's revenue before customer incentives, divided by transaction value. |
| Incentive rate | s | Customer incentives divided by transaction value. |
| Invisible wedge | W | Transaction value minus revenue: what a revenue figure leaves out. It is defined for one platform. The thesis sums it over three platforms in Indonesia for 2023 and computes no national total. |
| Ecosystem ratio | E | The wedge divided by revenue. It equals one divided by the take rate, minus one. This is the author's own measure, built for this thesis. |
| Growth | g | The change from one year to the next as a share of the first year: 0.09 means 9 percent. |
| Growth divergence | D | Growth of revenue minus growth of transaction value in the same year: 0.62 means 62 percentage points. |
| Online seller | | A business that the national statistics office, BPS, counts as selling online; the text also calls it a seller or a business. The count of online sellers is BPS's estimate of how many there are in a year. |
| First-time entry | | A seller that began selling online in the year being measured. |

Three rules decide which figures can be set against each other. First, transaction value and revenue are paired only when they refer to the same period and the same business. A take rate built from figures of different scope would measure the mismatch, not the platform.

Second, each platform is compared only with itself from one year to the next. Companies define transaction value in their own ways, and a disclosed measure tells less when the way it is computed changes (Givoly, Li, Lourie and Nekrasov 2019). Within one company the definition cancels out of the yearly change, as long as the company keeps it.

Third, companies that mainly sell goods themselves are kept out. Such a company reports the whole sale as revenue, while a platform that only arranges the sale reports its fee (International Accounting Standards Board 2014). The two business models differ in kind (Hagiu and Wright 2015), and their take

rates cannot be compared.

Every figure also keeps a label for where it comes from. Figures that companies report are kept apart from figures that the thesis constructs, and both are kept apart from national statistics and payment figures. Chapter 4 applies these rules platform by platform.

## 3.2 Three measures

The invisible wedge measures, in currency, the part of transaction value that lies outside a platform's revenue:

$$W_{it} = V_{it} - R_{it} \tag{1}$$

where $V_{it}$ is the transaction value of platform i in year t and $R_{it}$ is the revenue it recognizes for the same year and business. The wedge is not an accounting error, because both amounts may appear in the same company report. It is a measure of transaction value, not of income or value added, so it does not measure missing output.

To compare platforms of different size, the thesis also expresses the wedge relative to revenue, as the ecosystem ratio:

$$E_{it} = W_{it} / R_{it} = 1 / m_{it} - 1 \tag{2}$$

where $m_{it} = R_{it} / V_{it}$ is the take rate, the share of each sale that the platform keeps. The ecosystem ratio is the author's own measure, and it restates the take rate. An ecosystem ratio of 9 means that 9 units of transaction value lie outside revenue for every unit recorded as revenue. Equivalently, the platform keeps 10 percent of each sale.

A single ratio does not show whether the relationship holds over time, so the thesis defines growth divergence:

$$D_{it} = g(R_{it}) - g(V_{it}) \tag{3}$$

where g(X) is the growth of X from one year to the next, as a share of the first year. A positive D means that revenue grew faster than transaction value, so the take rate rose. A negative D means the reverse. If revenue were a stable proxy for transaction value, D would stay close to zero. Revenue equals the take rate times transaction value, so D can also be written with the take rate alone:

$$D_{it} = (1 + g(V_{it})) \times (m_{it} - m_{i,t-1}) / m_{i,t-1} \tag{4}$$

where $m_{i,t-1}$ is the take rate in the earlier year and $m_{it}$ the take rate in the later year. If the take rate is steady, D is zero and revenue grows exactly as fast as transaction value.

Tokopedia between 2022 and 2023 illustrates the three measures. Its take rate rose from about 1.5 to about 2.5 percent, so it kept about two thirds more of each sale. Its transaction value fell by about 9 percent and its revenue rose by about 53 percent, so D was $0.91 \times 0.68 = 0.62$, or 62 percentage points. Its wedge stayed at about 98 percent of transaction value and fell with it, by about 10 percent.

The three measures do different jobs. The wedge gives the size of what revenue leaves out, and at a marketplace that keeps a few percent of each sale it is almost all of transaction value. The ecosystem ratio expresses that size relative to revenue. Growth divergence carries the central test: whether revenue and transaction value keep the same relationship from year to year.

The ecosystem ratio also governs how far a change in the take rate moves revenue growth. Equation (4) shows that D is the change in the take rate divided by its starting level.

A platform that keeps 2 percent of each sale has an ecosystem ratio of 49. If its transaction value is flat, a rise in its take rate to 3 percent raises its revenue by 50 percent. A platform that keeps 17 percent has an ecosystem ratio near 5, and a rise to 18 percent raises its revenue by about 6 percent. So a large ecosystem ratio means that ordinary changes in fees or discounts can produce large swings in reported revenue growth.

## 3.3 Tracing a movement

A large divergence calls for an explanation, and the same definitions supply it. Measured in logarithms, the growth of revenue splits exactly into the growth of transaction value and the change in the take rate:

$$\Delta \ln R_{it} = \Delta \ln V_{it} + \Delta \ln m_{it} \tag{5}$$

where $\Delta \ln X$ is the change in the natural logarithm of X from one year to the next. The first term is the volume effect, more or less commerce. The second is the take-rate effect, a larger or smaller share of each sale. A log change treats a rise and a fall by the same factor as equal in size, so the two effects can be compared directly. The take rate itself has two parts:

$$m_{it} = f_{it} - s_{it} \tag{6}$$

where $f_{it}$ is the fee rate and $s_{it}$ the incentive rate, both measured against transaction value. A take rate can rise because a platform charges more or because it spends less on discounts and vouchers. Accounting rules normally deduct such incentives from revenue (International Accounting Standards Board 2014). So a cut in incentives raises revenue even when no fee changes.

Equations (5) and (6) together divide a movement in revenue into more commerce, higher fees and lower incentives. A change in the mix of a company's businesses is a further source, read from its reports where they describe one. The split shows how a movement happened. It does not show why the company chose it, and company explanations are read as reports.

Equation (6) also shows where a rule acts. A cap on commissions acts on the fee rate, and a limit on discounts acts on the incentive rate. Chapter 6 sets Indonesia's recent rules against these parts of the take rate.

## 3.4 Hypotheses

Two hypotheses follow from the research question. H1 asks whether revenue follows the commerce carried within a platform. H2 asks where national online growth comes from, which decides whether the commerce on platforms can stand for national online commerce. Together they test the two links in Figure 3.1. Table 3.2 pairs each hypothesis, and the entry test of Section 3.5, with its main test and with the result that would count against it.

**Table 3.2. Hypotheses, tests and what would count against them**

| Test | What it asks | Main test | What would count against it |
|---|---|---|---|
| H1 | Do revenue and transaction value grow together within a platform? | Size of yearly changes in the take rate, Indonesian and regional platforms against listed platforms abroad | Changes in Indonesia far larger than abroad |
| H2 | Does national e-commerce value grow because of more businesses or because of higher sales per business? | BPS's national e-commerce value split into the number of businesses and sales per business | Higher sales per business accounting for most of the growth |
| Entry test | Does a rise in the count of online sellers come from first-time entry? | Share of 2023 sellers in BPS's survey files who began selling online in 2023 | Fewer new sellers than the rise requires |

The share of transaction value that a platform keeps is not fixed. A platform sets its charges on both sides of the market together, and both the level and the split are choices (Rochet and Tirole 2003). Platforms often hold down the price to one side to draw it in. They can raise that price once both sides have joined (Evans and Schmalensee 2016).

Accounting rules move the share in two ways. The discounts a platform pays for come out of its revenue. A platform that sells goods itself, a different business model (Hagiu and Wright 2015), records more of each sale than one that only arranges it (International Accounting Standards Board 2014). Both can change as a platform changes its business model or scope.

Revenue follows transaction value only while the take rate holds steady, that is, while D stays near zero. In that case revenue would be a workable proxy for the commerce a platform carries, and the ecosystem ratio would not change. The first hypothesis states this case.

> **H1.** Transaction value and revenue recognized by the same platform grow in proportion over time.

Growth rates never match exactly, so H1 is tested by comparison. The test compares the size of yearly changes in the take rate, measured in logs as in equation (5), between two groups. One group is the Indonesian and regional platforms. The other is listed platforms abroad, where revenue and transaction value broadly move together. Changes far larger in Indonesia would count against H1. Chapter 4 describes the two groups and the test.

National e-commerce value can grow in two ways. Existing businesses can sell more, which is the intensive margin. More businesses can sell online, which is the extensive margin. Trade economists split a country's exports in the same way, into more of each good and more kinds of goods (Hummels and Klenow 2005).

The distinction matters for measurement. Growth driven by new sellers can take place in channels that platform accounts do not cover, such as social media and instant messaging, which BPS reports are widely used (BPS 2025a). Platform figures can then miss that growth even when every platform reports accurately.

BPS publishes both the number of businesses selling online, N, and their total online sales value, Y. Since Y equals N times the sales per business, its growth in logarithms splits exactly in two:

$$\Delta \ln Y_t = \Delta \ln N_t + \Delta \ln (Y_t / N_t) \tag{7}$$

where the first term is the extensive margin and the second the intensive margin. H2 holds if the first term is more than half of the total. The second hypothesis states the case in which more businesses drive growth.

> **H2.** The growth of national e-commerce value comes mainly from more businesses selling online, and less from higher sales per business.

The H2 split is arithmetic. It does not assign new businesses to channels, and it does not show whether the businesses in the count are new.

## 3.5 The entry test

H2 rests on BPS's count of businesses selling online, and a rise in that count can mean two different things. Either more businesses started selling, or the survey reached businesses that were already selling. A change in how a survey finds its respondents can create a break in a series that looks like real change (van den Brakel, Smith and Compton 2008).

The published figures cannot tell the two apart. The survey can, because it asks every seller in which year it began selling online. The thesis checks what the count contains with a test written down before the survey files were opened.

> **Entry test.** In BPS's survey of online sellers for 2023, at least 21.5 percent of online sellers must report that they began selling online in 2023.

The 21.5 percent threshold comes from arithmetic. The published count of online sellers rose by 27.4 percent in 2023. Suppose every extra seller was new and no seller stopped. Then for every 100 sellers a year earlier there were 127.4, and 27.4 of those were new. New sellers would make up 21.5 percent of all sellers, which is 27.4 divided by 127.4.

Sellers who stopped would have to be replaced as well, which only raises the share needed. The threshold is the smallest share of new sellers that can produce the rise. The test does not test H2 itself. It tests what "more businesses" means.

Sellers' memory works in favour of passing the test. People tend to report past events as more recent than they were (Neter and Waksberg 1964). A seller who began in 2022 but reports 2023 raises the share of new sellers. Recall error of this kind makes the threshold easier to reach, so a share below it is a cautious finding.

If the share of new sellers falls short, part of the rise is something other than first-time entry. Two readings are open. The survey may have reached more of the sellers who were already there. Or sellers may have come back after a pause, since a returning seller keeps an old start year. Small firms that stop are more often young firms with young owners (McKenzie and Paffhausen 2019). So the age of the sellers behind the rise bears on the second reading. Chapter 5 weighs the two readings with that evidence.

## 3.6 Further analyses

Four further analyses complete the framework without forming hypotheses of their own. Each answers a part of the research question that H1, H2 and the entry test leave open. Table 3.3 lists them.

**Table 3.3. Further analyses**

| Analysis | What it asks | Main evidence |
|---|---|---|
| Tracing large movements | What moved revenue when it parted from transaction value? | Equations (5) and (6), applied to company reports |
| Market values | Do markets value growth from more commerce differently from growth in the take rate? | Year-end market values of listed platforms |
| Channels and reach | How much online selling passes through marketplaces, and how far can rules built on platform records reach? | BPS's channel shares and survey files |
| Comparing national figures | Do national figures tell the same story of growth? | BPS, Bank Indonesia, industry estimates and household spending |

Tracing large movements applies the split of Section 3.3. Wherever H1 finds a large change in the take rate, the thesis divides the change in revenue into its parts. It also reads the company's reports for changes in the mix of businesses and in reporting scope.

The market-value analysis asks whether investors, who price revenue (Section 2.1), treat its two sources alike. Equation (5) shows that revenue can grow in two ways, through more commerce or through a larger take rate. Early studies of internet companies found that markets priced measures of web activity beyond reported earnings (Trueman, Wong and Zhang 2000; Rajgopal, Venkatachalam and Kotha 2003).

The two sources of growth need not be worth the same. Growth from the take rate has a limit, because a platform cannot raise its share of each sale without end. A higher price to one side of a platform can also reduce how much that side uses it (Rochet and Tirole 2003).

The thesis relates the yearly change in a platform's market value, MV, to the two sources of revenue growth:

$$\Delta \ln MV_{it} = \beta_V \, \Delta \ln V_{it} + \beta_m \, \Delta \ln m_{it} + \text{year effect} + \text{error} \qquad (8)$$

where $\beta_V$ shows how market value moves with transaction value, and $\beta_m$ how it moves with the take rate. Two predictions were written down before the market values were collected. The first is that market value moves with transaction value, so $\beta_V$ is positive. The second is that markets value growth from more commerce more than growth from a higher take rate, so $\beta_V$ is larger than $\beta_m$.

The channel analysis answers the third question of Section 2.4. A rule built on platform records reaches only the sellers whose sales pass through a platform. Research on tax shows that reports from a third party improve compliance where they exist (Kleven et al. 2011).

The reach of a rule built on platform records depends on how online selling divides between marketplaces and channels that no marketplace records, such as messaging apps and social media. The thesis measures that division by sellers and by sales value. It also counts the marketplace sellers who may be large enough to fall under the marketplace tax.

Record keeping is the second part of this analysis. The thesis compares how often marketplace sellers and other sellers keep financial statements. The comparison bears on the information that platform records can add.

The comparison of national figures extends the outer link in Figure 3.1. Platform reports, BPS's survey, Bank Indonesia's statistics and industry estimates count different things. Their values change with the definition used (UNCTAD 2024).

If two national figures move in opposite directions in one year, the conclusion about growth depends on the figure chosen. The thesis compares growth rates, each in its own unit, and levels only after reconciling what each source covers. Household spending is a further point of comparison.

# Chapter 4. Data and Methods

The figures this thesis compares come from sources that were never designed to be read together. Platforms publish in their own formats, and the answers of individual sellers are released only under a data-use agreement. The methods in this chapter are built around these conditions.

On the platform side, the thesis reads revenue and transaction value from annual reports, audited statements and earnings releases. The extraction tables keep the source file and the quoted passage beside every value. The Indonesian platforms are three online marketplaces, Tokopedia, Bukalapak and Blibli, and GoTo, the group that runs the Gojek app for rides and deliveries. GoTo also controlled Tokopedia until 2024 and still holds a stake in it. Two regional companies, both based in Singapore, complete the group: Grab, which runs rides and deliveries across Southeast Asia, and Sea, whose marketplace is Shopee. Where a platform does not report Indonesia alone, as with Shopee and Grab, the thesis states what it derives or borrows and labels it.

The national side rests on BPS's survey of online sellers. BPS publishes a few totals, and the thesis obtained the answers behind them for three survey rounds, which show when each seller began selling online and how much it sold. The central bank's own figure has no published method, so it serves only as a point of comparison.

Setting the two sides against each other for one country is what makes the tests possible. Indonesia allows it. Five platform series report revenue and transaction value in the same year, and three of them report what they pay out in incentives. The survey asks every online seller when it began. The tests are built to ask only what these sources can answer, and each limit is stated where it applies.

The chapter describes the evidence on each side and the tests that use it. It starts with the platforms: the data, the wedge in Indonesia, the test of H1, the tracing of large moves and market values. It then turns to the survey: H2, the entry test, and the channels and records of sellers. It ends with the comparison of national figures and with how every number is labelled and traced.

## 4.1 The platform data

The basic observation is one platform in one year, which this thesis calls a platform-year. Revenue comes from the company's financial statements and transaction value from its annual report or earnings release. Both refer to the same year and, as closely as the filings allow, the same business.

Companies define transaction value in different ways, and some include shipping charges, digital goods or travel. The thesis keeps each company's own definition and records it. Under the second rule of Section 3.1, the definition cancels out of a take rate's yearly changes. This holds as long as the company keeps both the definition and the mix of its businesses.

Two groups of platforms are compared. The Indonesian and regional group has five platforms, listed in Table 4.1. Three are Indonesian marketplaces, and two are the on-demand businesses of Grab and GoTo, which cover rides and deliveries. The foreign group has 27 listed platforms in marketplaces, travel and ticketing, freelance work, resale and food delivery. Eight were chosen first, and 19 were added to widen the comparison. Ten more were examined and left out, six because they sell mostly their own goods and four because they do not disclose transaction value.

**Table 4.1. The Indonesian and regional group of platforms**

| Platform series | What it covers | Years | Yearly pairs | How it is labelled |
|---|---|---|---|---|
| Tokopedia | The e-commerce segment of GoTo, an Indonesian marketplace | 2022–2023 | 1 | Direct segment disclosure |
| Blibli | Blibli's third-party retail segment, which includes online travel | 2021–2025 | 4 | Direct segment disclosure |
| Bukalapak | The whole Bukalapak group, mainly in Indonesia, including Mitra, its online-to-offline business | 2020–2023 | 3 | Direct company disclosure |
| Grab on-demand | Grab's rides and deliveries across Southeast Asia, not Indonesia alone | 2020–2024 | 4 | Direct segment disclosure |
| GoTo on-demand | GoTo's rides and deliveries segment, in Indonesia and Vietnam to 2024 | 2022–2025 | 3 | Direct segment disclosure; the 2022 transaction value is an estimate |

Table 4.1 states the scope of each series because the series differ, and only Tokopedia is an Indonesian segment alone. The thesis never adds the series together, and each platform enters the tests on its own. Sea, whose Shopee business has Indonesia as a main market, sits in the foreign group.

Platform-years are compared in pairs, and three rules keep the pairs comparable. Two consecutive platform-years form a pair, and nothing else does. A missing year stays missing and is never filled in. Annual and quarterly pairs are never pooled. A take rate is a ratio of two amounts in one currency, so the tests need no exchange rate. Amounts in running text are in US dollars at the World Bank's average rate for the year.

The third rule of Section 3.1, which keeps out companies that mainly sell their own goods, is applied in two ways. For the 19 platforms added, the thesis drops any year in which own goods exceed half of revenue. For the eight chosen first, it keeps a pair only when the company's reporting scope did not change between the two years.

Two platforms sit close to that line. Zalando is one of the eight and sells part of what it carries, so Appendix B repeats the test without it. One added pair, Ozon from 2021 to 2022, ends in a kept year but starts in a dropped one. Appendix B repeats the test without Ozon as well. The result is 15 pairs for the Indonesian and regional group and 158 for the foreign group.

## 4.2 Measuring the wedge in Indonesia

The first objective, the size of the invisible wedge, needs amounts for Indonesia alone. The thesis sums the transaction value and revenue of three major platforms in Indonesia in 2023: Tokopedia, Shopee and Grab. The wedge and the ecosystem ratio are then computed from the sums.

Each platform reaches the sum by its own route. Tokopedia reports both amounts for its segment. Shopee's transaction value is an outside estimate by Momentum Works, a research firm that tracks online commerce in Southeast Asia. Its revenue assumes a take rate of 10 percent, which the thesis varies from 9 to 11 percent. Grab reports its Indonesian revenue, and the thesis derives its Indonesian transaction value from Grab's company-wide ratio of the two amounts. The thesis drops each platform in turn, and it repeats the exercise for every platform-year between 2020 and 2025 that the filings allow.

## 4.3 Testing H1

The test of H1 asks how far each platform's take rate moves from one year to the next. For each pair of years it takes the change in the logarithm of the take rate, the take-rate term of equation (5). This change is growth divergence measured in logs, and it does not depend on how fast transaction value grew. Tokopedia's log change in 2023 is 0.52, which is a rise by a factor of 1.68, or 68 percent.

A move in either direction counts against H1, so the test uses the size of each yearly change in the take rate. Each platform gets one number, the median size of its yearly changes, because the pairs of one platform are not independent of one another.

The test compares five numbers for the Indonesian and regional group with 27 for the foreign group. It asks whether the first group ranks higher than chance would give, using a one-sided Mann–Whitney test (Mann and Whitney 1947). It repeats the question by shuffling the group labels 9,999 times. The p-value is the chance of a result at least this lopsided if the groups did not differ.

Appendix B lists the checks on this test. They drop each platform in turn and treat Tokopedia and GoTo as one company. They repeat the test with growth divergence in points as Chapter 3 defines it. They compare the change in the take rate in percentage points, to see whether a thin take rate explains the result. They narrow the test to the three marketplaces, correlate revenue growth with transaction value growth, and repeat the test with quarterly pairs.

## 4.4 Tracing the moves

The second objective, to reconcile large movements in revenue with disclosed changes in fees, discounts and scope, is answered in two steps. The first divides each movement into volume and take rate with equation (5). The take-rate share is the size of the take-rate change divided by the sum of the sizes of both changes. The same split is made for the wedge.

The second step splits the take-rate change with equation (6). Some companies disclose both their gross fees and their incentives. For them, the thesis splits the growth of revenue over a window into three parts: more commerce, a change in the fee rate and a change in the incentive rate. A change in the mix of businesses is recorded from the company's reports.

The sources differ by company. Tokopedia's audited segment note gives gross revenue, incentives and net revenue, and the thesis checks them against the audited consolidated statements. For Tokopedia the thesis also reports the change line by line in rupiah, the usual way to read accounts, and says where the two views differ.

Grab is read from its annual filings and GoTo from its audited statements. Blibli does not report incentives, so its split rests on a proxy built from its releases. Seven windows are traced, and Appendix A lists them. The thesis also records where accounting treatment or reporting scope matters, such as the deduction of incentives from revenue and the travel business inside Blibli's retail segment. Company explanations are read as reports and are not tested.

## 4.5 Market values

The market-value analysis of Section 3.6 uses year-end market values from the financial data service LSEG, under the university licence. The main sample has 28 listed platforms and 138 platform-years, made up of Bukalapak and 27 platforms abroad, and it uses whole-company figures only. It is not the list of 27 used for H1. It adds Airbnb, Booking, DoorDash, Lyft and Uber, and it lacks Despegar, Mercado Libre, Ozon, Rakuten and Sea.

Equation (8) is estimated with a fixed effect for each year, which removes what all platforms share in that year. Standard errors are clustered by company, which allows for the repeated observations of one company. Each prediction names a direction, so both tests are one-sided. The test was written down before the market values were pulled. The result is an association, and market value also changes when a company issues shares, so the thesis does not call it a stock return. A second run adds five segment series and Mercado Libre as a robustness check. Grab, GoTo and Sea enter only there, because their reported transaction value covers only part of the business that the market value prices.

## 4.6 The survey files and H2

The national evidence comes from the survey of online sellers that BPS runs. BPS publishes the number of businesses selling online, their total online sales and the shares by channel (BPS 2025a, 2025b). The survey rounds are named for the survey year, but their questions refer to the year before. The thesis names a file by its round and a year by the year of activity, so the file for the 2024 round describes 2023. Each round lists businesses in sampled areas and updates its sampling frame, so a change in who is found can change the count.

H2 uses the published figures and the split of equation (7). BPS prints two counts of businesses for 2023, so the thesis computes the 2023 to 2024 split with both and reports both. The 2022 to 2023 split uses the count that the survey weights match.

The published figures cannot tell a new seller from one counted for the first time, but the survey files can. Table 4.2 lists them. Each row is a business that sold online in the activity year. Its survey weight shows how many businesses it represents, so the weights add up to the national count.

Two checks come before any test. The weighted total of each file must fall within 1 percent of the BPS count. The 2024 file must also reproduce two published shares of sellers who keep complete financial statements, 28.63 and 12.25 percent. All three files pass. The files carry no business identifier across rounds and no area variable below the province. So the rounds compare groups of sellers and not the same sellers over time.

**Table 4.2. The three BPS survey files**

| Survey round | Activity year | Rows | Weighted total | BPS count |
|---|---|---|---|---|
| E-Commerce 2021 | 2020 | 11,848 | 2,361,423 | 2,361,423 |
| E-Commerce 2023 | 2022 | 31,724 | 2,995,986 | 2,995,879* |
| E-Commerce 2024 | 2023 | 39,736 | 3,814,799 | 3,816,750 |

*Implied by the published count for 2023 and its published growth of 27.4 percent.

## 4.7 The entry test and the extra sellers

The entry test asks whether at least 21.5 percent of the 2023 sellers in the 2024 file began selling online in 2023, a threshold that Chapter 3 derives. BPS also prints a second count for 2023, but the survey weights match the first, so the threshold stays at 21.5 percent. The test was written down and dated in the project's code archive, called the repository, before the files were opened.

A failed entry test raises a second question: how much of the rise does first-time entry not explain? The sellers behind that part are called the extra sellers. A cohort is the group of sellers that began selling online by a given year. If two surveys count the same population, a cohort can only shrink between them, because sellers stop and no new seller can join it.

The cohort check compares the 2024 file's count of sellers who began before 2023 with the whole 2022 count. It does so twice. The first time it assumes that no seller stopped. The second time it assumes that sellers stopped at about 5 percent a year, the rate at which one cohort shrank between the 2020 and 2022 surveys.

Further checks ask where the extra sellers sit. They look across provinces and channels, using rank correlations, and at sellers' own reports of their sales. They use the sampling error that BPS publishes. They profile who the extra sellers are, with each group's count in 2023 compared with what its own yearly change in 2020 to 2022 would give. They look at part-year selling and test a scenario of returners.

The cohort check and the further checks were designed after the entry test failed, so Chapter 5 labels them exploratory. The exception is the median change in sellers' sales, which was registered. The files cannot rule out returners, because a seller who stopped and returned keeps an old start year.

## 4.8 Channels, records and reach

Three analyses beyond the entry test also use the survey files. One asks which channels sellers use and how sales divide between them. The published tables split the 2023 to 2024 rise in value between marketplaces and other channels, and BPS counts food and ride apps in its marketplace category.

Value by channel rests on revenue brackets, which only the 2021 and 2023 files give. The thesis assigns each bracket a low, a middle and a high amount. The amounts were fixed before the file was opened. A validation rule applies: the middle scenario's 2022 total must lie between half and double BPS's published total of about US$53 billion.

The 2023 questionnaire asks for monthly revenue but the bracket limits are annual size classes, so both readings are computed and the one that passes is kept. If the lower brackets alone already exceed the total, value shares are reported as ranges.

Whether a seller keeps complete financial statements is the subject of a second analysis. It bears on what platform records can add, because a seller with no statements leaves fewer formal records. World Bank Enterprise Surveys add a check on records. They cover unregistered firms in six Indonesian cities in 2023, and formal firms with five or more employees in seven countries.

A third analysis builds on the first two and is developed in Chapter 6. It counts the marketplace sellers whose revenue is above a bracket line and expresses the count as a share of all online sellers. At the 2022 rate, BPS's brackets divide at about US$20,000 a year. The marketplace tax exempts individual sellers up to about US$34,000. So the count above BPS's line is an upper limit on the sellers the tax can reach. Chapter 6 also sets each recent rule against the part of the figures it acts on. The thesis reads each legal text where it is published. It labels each rule by the status of its text: primary text read, announced, or reported in the news only.

## 4.9 Comparing national figures

The third objective, to compare the platform evidence with national and payment figures, rests on the sources that Table 4.3 lists, with what each one counts. Beyond the rules of Section 3.1, the comparison is closest where the coverage of two sources overlaps. Payment figures serve as context, because they show how people pay and not what they buy online.

The 2023 file, which describes 2022, lets the thesis separate the part of BPS's marketplace sales that went to final consumers. It compares that part with Bank Indonesia's figure for the same year, in a test written down before the file was opened. The thesis sets the 2023 growth of six measures of buying through platforms beside household spending, each in its own unit. It does the same with quarterly platform growth.

**Table 4.3. Sources compared in Objective 3**

| Source | What it covers | How it counts |
|---|---|---|
| Platform reports | Everything bought through one platform | The platform's own records |
| BPS online-seller survey | Every business that sold online, including chat and social media | Listing in sampled areas, scaled up by survey weights |
| Bank Indonesia e-commerce statistics | Transactions of large e-commerce firms (Bank Indonesia 2025) | Data sent to the central bank; the method is not public |
| Industry estimates | Shopping marketplaces in Indonesia or Southeast Asia, from the research firm Momentum Works (2024) and the e-Conomy report (Google, Temasek and Bain 2024) | Outside estimates |
| Household spending | All household purchases | BPS national accounts |
| Payment figures | Payments made through systems such as QRIS, the national QR-code system | Payment records, in person and online |
| Adjacent figures | Parcels shipped in Southeast Asia, the value-added tax collected on foreign digital services (Directorate General of Taxes n.d.), and the number of merchants a company reports | Used as context only, since none measures Indonesian e-commerce |

## 4.10 Labels, traceability and licensed data

Every number in the thesis follows one path: source document, extract, analytic row, result. Each extract keeps the company's currency and units, and each value carries a label: direct company disclosure, direct segment disclosure, external estimate, derived value, conversion or scenario. The labels are never mixed in one sample. A restatement or a conflict between two reports is reported and never overwritten. A single script rebuilds every table that depends only on committed files, and 272 automatic checks confirm that each recorded result still matches its table.

The thesis also separates tests by when they were written. A registered test was written down, dated and stored in the repository before the data were opened. This applies to the entry test, the scenario rules for value, the comparison with Bank Indonesia and the market-value test, among others. Appendix C lists them with their dates. An exploratory test was designed after some results were seen. The test with the larger groups in Section 4.3 belongs here, because it was set up after a first test on the three marketplaces. A descriptive result reports numbers without a test.

Three sources are licensed and are not in the repository: the BPS survey files, the LSEG market values and the World Bank survey files. The repository records the size and checksum of each BPS and LSEG file and of the World Bank cross-country query, so anyone who obtains the same files can check them. It holds the code that reads them and the national totals that the code produces, and no business-level record. BPS is cited as the source of every survey figure, in line with the data-use agreement. Chapter 7 states what these data cannot show.

# Chapter 5. Results

The results follow the two links of Figure 3.1. On the inner link, revenue showed a small and unsteady part of the commerce that platforms carried. Platforms changed how much of each sale they kept, and revenue moved with those choices (Sections 5.1 to 5.4). On the outer link, the national count of online sellers rose by 27 percent in 2023. About half of that rise is not explained by sellers entering for the first time, and whether H2 holds depends on how that half is read (Sections 5.5 to 5.10).

Every result in this chapter carries one of the three labels of Section 4.10: registered, exploratory or descriptive. A result given without a label is descriptive. Table 5.1 gives each main result with its label, the checks behind it and the section that reports it.

**Table 5.1. Main results and the evidence behind them**

| Result | Label | Main checks | Section |
|---|---|---|---|
| *Inner link: revenue and platform commerce* | | | |
| Revenue was about 7 percent of what three major platforms carried in 2023; the wedge was about US$40 billion | Descriptive | Shopee's take rate varied; each platform dropped in turn | 5.1 |
| Take rates moved about three times as much in Indonesia and the region as abroad, so H1 is rejected | Exploratory; the narrower test, set before the wider one, points the same way | Each platform dropped in turn; Ozon and Zalando dropped; change measured in points; quarterly pairs do not reject | 5.2 |
| The thin take rate of Indonesian and regional platforms explains much of that difference | Exploratory | Changes in percentage points compared across groups | 5.2 |
| Pricing, not more commerce, moved revenue | Descriptive | All 15 pairs against 158 abroad; seven traced windows; two views of Tokopedia | 5.3 |
| Market value moves with transaction value | Registered; holds | Extreme values capped (winsorised); a second run | 5.4 |
| Markets value growth from a higher take rate less | Registered; does not hold in the main sample | Holds only in a second run that mixes segment figures | 5.4 |
| *Outer link: platform commerce and all online commerce* | | | |
| Buying through platforms, and sellers' own sales, did not boom in 2023 | Descriptive; registered for sellers' sales, holds | Six platform measures against household spending | 5.5 |
| More businesses account for most national growth, so H2 holds | Descriptive | Both BPS counts for 2023; two periods | 5.6 |
| Too few 2023 sellers were new to produce the rise in the count (13.45 against 21.5 percent) | Registered; the entry test fails | Recall error would favour passing | 5.6 |
| About half of the rise is not explained by first-time entry | Exploratory | With and without exits; cohorts; provinces; sampling error | 5.7 |
| Read as wider coverage, the extra count leaves H2 near its threshold (about 42 to 54 percent) | Exploratory | Count and sales removed together; two exit rates | 5.7 |

| Result | Label | Main checks | Section |
|---|---|---|---|
| The extra count leaned toward long-running businesses; returning sellers cannot be ruled out | Exploratory | Part-year selling | 5.8 |
| Most online selling happens outside marketplaces, and records are thinner there | Registered and descriptive | Four survey years; World Bank surveys | 5.9 |
| National figures agree only where they cover the same sellers | Registered, with a mixed outcome, and descriptive | Three value scenarios | 5.10 |

## 5.1 How much revenue leaves out

Revenue showed only a small part of what Indonesia's major platforms carried. In 2023, Tokopedia, Shopee and Grab together carried about US$43 billion of transaction value and kept about US$3 billion as revenue. Revenue was about 7 percent of that commerce, and the invisible wedge was about US$40 billion. The ecosystem ratio was about 13, so almost 13 dollars of commerce passed outside revenue for every dollar the platforms kept. Table 5.2 shows each platform.

**Table 5.2. The invisible wedge of three platforms, 2023 (US$ billion)**

| Platform | Transaction value | Revenue | Wedge | Ecosystem ratio | How it is labelled |
|---|---|---|---|---|---|
| Tokopedia | 16.33 | 0.41 | 15.93 | 39.3 | Both direct segment disclosure |
| Shopee, Indonesia | 21.52 | 2.15 | 19.37 | 9.0 | Transaction value is an outside estimate (Momentum Works); revenue is derived at an assumed take rate of 10 percent |
| Grab, Indonesia | 5.38 | 0.60 | 4.78 | 7.9 | Revenue is a direct country disclosure; transaction value is derived from Grab's company-wide ratio |
| Total | 43.23 | 3.16 | 40.07 | 12.7 | Sum of three platforms, not a national total |

*Note:* Rounded amounts may not add. With Shopee's take rate at 9 or 11 percent, the total wedge is US$40.3 or US$39.9 billion; at Sea's company-wide rate of 11.5 percent, it is US$39.8 billion. Source: company filings; Momentum Works (2024); author's calculations.

The ratios differ because the amounts were obtained in different ways. Tokopedia reports both amounts, and its ratio is about 39. Shopee's ratio of 9 follows from the assumed take rate, and Grab's ratio of about 8 is its company-wide ratio. Neither is separate evidence for Indonesia. Blibli and Bukalapak also report both amounts, with ratios of about 43 and 36, close to Tokopedia's. They are left out of the sum because Blibli's figure includes online travel and Bukalapak reports for its whole group.

The size of the wedge does not depend on the assumed rates. A take rate for Shopee between 9 and 11 percent keeps the total near US$40 billion. Dropping any one platform still leaves more than US$20 billion. For platforms that report both amounts, the wedge was 97 to 99 percent of transaction value in every year. It moved almost one for one with the commerce, and Appendix A gives the wedge by platform and year.

Platform commerce is itself only part of online commerce. In BPS's survey, marketplaces carry about a fifth of online sales value (Section 5.9), so platform revenue shows a smaller part still. A small share can still be a fair guide to the commerce, as long as it keeps its size from year to year.

## 5.2 Testing H1

Revenue did not keep pace with transaction value at the Indonesian and regional platforms, and the thesis rejects H1. At the typical platform in this group, the take rate moved by about 20 percent a year, up or down. At the typical foreign platform it moved by about 6 percent, so the moves were about three times as large. Table 5.3 gives the test and its checks.

**Table 5.3. The test of H1**

| Test | Method | Indonesian and regional group | Foreign group | p-value |
|---|---|---|---|---|
| Yearly change in log take rate, 5 platforms against 27 | Mann–Whitney | 0.20 | 0.06 | 0.004 |
| The same | Shuffle | 0.20 | 0.06 | 0.022 |
| Growth divergence in points, 5 platforms against 26 | Mann–Whitney | 27.0 | 6.5 | 0.005 |
| Drop each platform in turn, or treat Tokopedia and GoTo as one company | Mann–Whitney | 0.17 to 0.23 | 0.06 | 0.003 to 0.016 |
| Quarterly pairs, 3 platforms against 8 | Mann–Whitney | 0.071 | 0.049 | 0.19 |

*Note:* Group values are medians of each platform's median yearly change. Appendix B lists every check, including the tests without Ozon and Zalando.

The main test compares the five platforms with 27 abroad, one median per platform. Both methods of Section 4.3 reject equal moves, with $p = 0.004$ and $p = 0.022$. No single platform carries the result. Dropping any one, or treating Tokopedia and GoTo as one company, keeps the Mann–Whitney p-value at 0.016 or below. The main test was set up after a first, narrower test, so it is exploratory.

A narrower test, set before the wider one, compares the three Indonesian marketplaces with the eight foreign platforms chosen first. It points the same way. With one median per platform, the Mann–Whitney test rejects equal moves ($p = 0.024$). A shuffle of platform labels on pairs of years comes close without passing ($p = 0.06$), since three platforms give a test little power. In that sample, revenue growth and transaction value growth had a correlation of 0.10 in Indonesia and 0.98 abroad, excluding Sea.

The direction of the moves is less telling than their size. Revenue grew faster than transaction value in 13 of the 15 Indonesian and regional pairs. In two of them, revenue and transaction value moved in opposite directions. Abroad, take rates also rose in about two thirds of pairs, but by much less. What sets the Indonesian and regional platforms apart is the size of the moves, which is what the test measures.

Quarterly pairs do not reject H1 ($p = 0.19$), contrary to a prediction registered before the test was run. The quarterly Indonesian pairs mostly cover 2023 to 2026, after most of the repricing, while the large moves came between 2022 and 2024. The rejection of H1 describes that period.

The large moves were not simply a mark of young platforms, an exploratory check. The Indonesian and regional platforms made their largest moves one or two years after listing on the stock market. Foreign platforms, by contrast, did not move their take rates more in their first three years after listing than later (median 0.07 against 0.08, $p = 0.24$). Nor did foreign platforms in lower-income markets move more in 2022 to 2024: one of seven did, against all four Indonesian and regional platforms with enough years. The repricing was an Indonesian and regional episode of those years.

The thin take rate of Indonesian and regional platforms explains much of the threefold difference, an exploratory finding. Measured in percentage points, both groups changed their take rates by about 1 point a year. The difference between them is nowhere near large enough to count ($p = 0.65$).

The two groups differ in the take rate they start from. The typical Indonesian and regional platform keeps about 1.6 percent of each sale, against about 17 percent abroad. The thinnest take rates belong to the three marketplaces, while Grab and GoTo keep shares closer to those abroad. As Section 3.2 showed, the same 1-point rise then has very different effects. It lifts revenue by more than half at the thinner take rate and by about 6 percent at the thicker one. Figure 5.1 shows the three measures side by side. The test shows that take rates moved, but not what the platforms changed to move them.

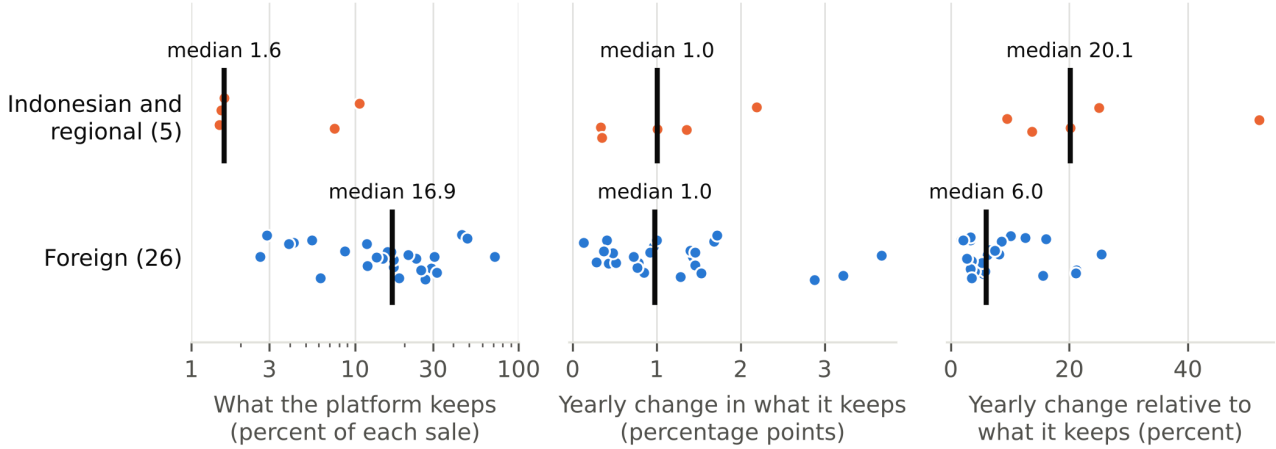



**Figure 5.1. The thin take rate.** Each dot is one platform's median over its years. Left: the share of each sale that the platform keeps, on a log scale. Middle: the yearly change in that share, in percentage points. Right: the same change relative to the share itself, as a log change in percent. Black bars mark group medians. The foreign group has 26 platforms here and 27 in the main test, because one platform has no pair of usable years for the change in points. Source: company filings; author's calculations. Exploratory.

## 5.3 What moved the take rate

Pricing, not more commerce, moved revenue at the Indonesian and regional platforms. Pricing here means a change in the take rate, through fees or incentives. With equation (5), the take rate accounts for about two thirds of a revenue move at the typical pair, against about a quarter abroad. The wedge, by contrast, follows the commerce: in both groups, about 95 percent of a move in the wedge comes from transaction value.

Where companies disclose both their fees and their incentives, equation (6) splits the growth of revenue further. Table 5.4 gives the seven traced windows. Fees and incentives are measured per unit of transaction value, and a negative share means that a part pulled revenue down. The windows of one platform overlap, so they are not seven independent cases.

**Table 5.4. What moved revenue in seven Indonesian and regional windows (share of the growth of revenue, percent)**

| Platform and window | Growth of revenue | More commerce | Higher fees | Lower incentives |
|---|---|---|---|---|
| Tokopedia, 2022–23 | +53% | −22 | 76 | 45 |
| Blibli third-party retail, 2022–23 | +465% | 17 | 20 | 63 |
| Blibli third-party retail, 2022–25 | +679% | 18 | 63 | 20 |
| Grab on-demand, 2021–24 | +321% | 34 | 34 | 33 |
| Grab on-demand, 2022–24 | +86% | 42 | 11 | 47 |
| GoTo on-demand, 2022–25 | +111% | 12 | 36 | 52 |
| GoTo on-demand, 2023–25 | +114% | 27 | 6 | 67 |

*Note:* Shares are rounded, so a row can add to 99 or 101. Source: company filings; author's calculations (Appendix A).

Higher fees and lower incentives together outweighed more commerce in every window. Lower incentives were the larger of the two in four windows and higher fees in two. In the seventh, Grab from 2021 to 2024, the two were nearly equal. Three series need a note. GoTo's 2022 transaction value is an estimate. Blibli does not report incentives, so its split rests on a proxy, and part of its 2022–23 rise comes from travel and digital products. Bukalapak's reports do not allow the parts to be separated.

Tokopedia can be read in two ways, and both point to pricing. Its incentives fell by about a third in 2023. Counted line by line in its accounts, lower incentives gave about 60 percent of the extra revenue and higher gross fees about 40 percent. That count does not adjust for the fall in transaction value, which lowered both lines. Measured per unit of transaction value, as in Table 5.4, higher fees account for more of the rise in the take rate. The two views differ only on which lever led (Figure 5.2).

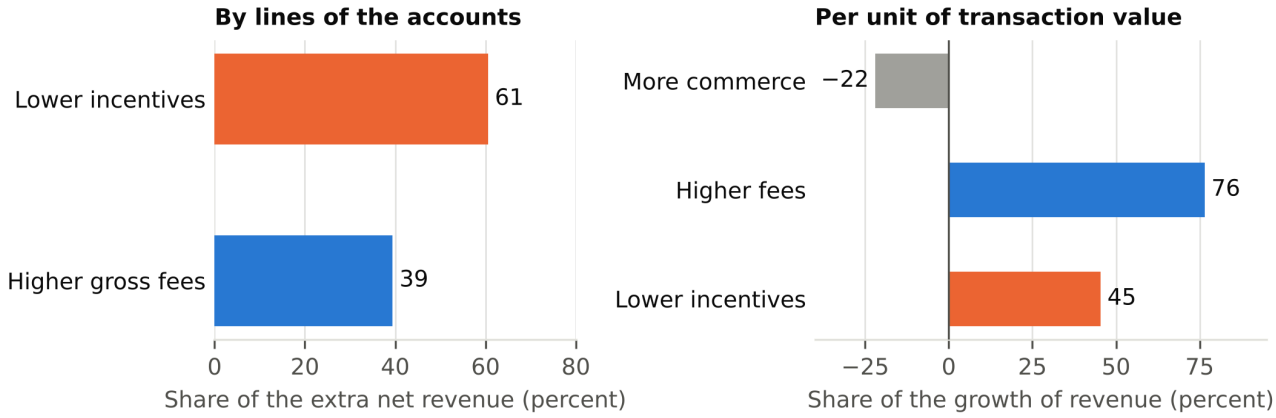



**Figure 5.2. Tokopedia, 2022 to 2023: two ways to split the rise in revenue.** Left: the change in net revenue split by the lines of the accounts. Right: the growth of revenue split per unit of transaction value, where a negative share pulled revenue down. Source: GoTo annual report 2023, segment note; author's calculations.

Tokopedia is not the only case of revenue rising while the commerce shrank. From 2024 to 2025, Blibli's transaction value fell by about 2 percent while its revenue rose by about 12 percent. In the windows traced here, revenue growth was not a fair guide to the commerce the platforms carried. Whether markets treated it as one is a separate question.

## 5.4 What markets valued

Across 28 listed platforms and 138 platform-years, market value moved with transaction value, a registered result. A 1 percent rise in transaction value goes with a rise of about 0.6 percent in market value ($\beta_V$ in equation (8), $p = 0.002$). Section 4.5 describes this sample, which differs from the 27 foreign platforms of H1.

The second registered prediction does not hold in the main sample. It expected markets to value growth from a higher take rate less than growth from more commerce. A 1 percent rise in the take rate goes with a rise of about 0.3 percent ($\beta_m$). That estimate is too uncertain to tell from zero. The gap between the two estimates is not large enough to count ($p = 0.20$), or $p = 0.16$ with extreme values capped.

A second run adds five segment series, including Grab, GoTo and Sea, and Mercado Libre. There the gap counts ($p = 0.004$), with about 0.6 for transaction value against about 0.1 for the take rate. That run sets segment figures against whole-company market values, as the registration warned, so the thesis does not rely on it. The answer depends on which platforms are in the sample.

Market value follows the commerce a platform carries. Early studies found a similar link, with markets pricing measures of web activity beyond earnings (Trueman, Wong and Zhang 2000; Rajgopal, Venkatachalam and Kotha 2003). Here the measure of activity is transaction value, which the platforms report themselves. How a firm records revenue matters too. After the 2000 crash, revenue explained less of the prices of internet firms that booked the full sale amount as revenue (Davis 2002). The data do not show that markets valued growth from pricing any less. Revenue and transaction value come from the platforms' own reports, and market value from share prices. The national figures give a separate view of the same commerce.

## 5.5 The platforms and the sellers in 2023

Buying through platforms did not boom in 2023. Household spending rose by about 9 percent in current prices that year. Of six measures of buying through platforms, five grew more slowly, and three fell. Blibli is the one exception. Figure 5.3 sets the measures side by side, each in its own unit.

The figures that described a boom were of other kinds. Tokopedia's revenue rose by 53 percent and BPS's online sales value by 41 percent, while the buying that platforms carried grew slowly or fell. Figure 5.3 shows the two kinds in different colours.

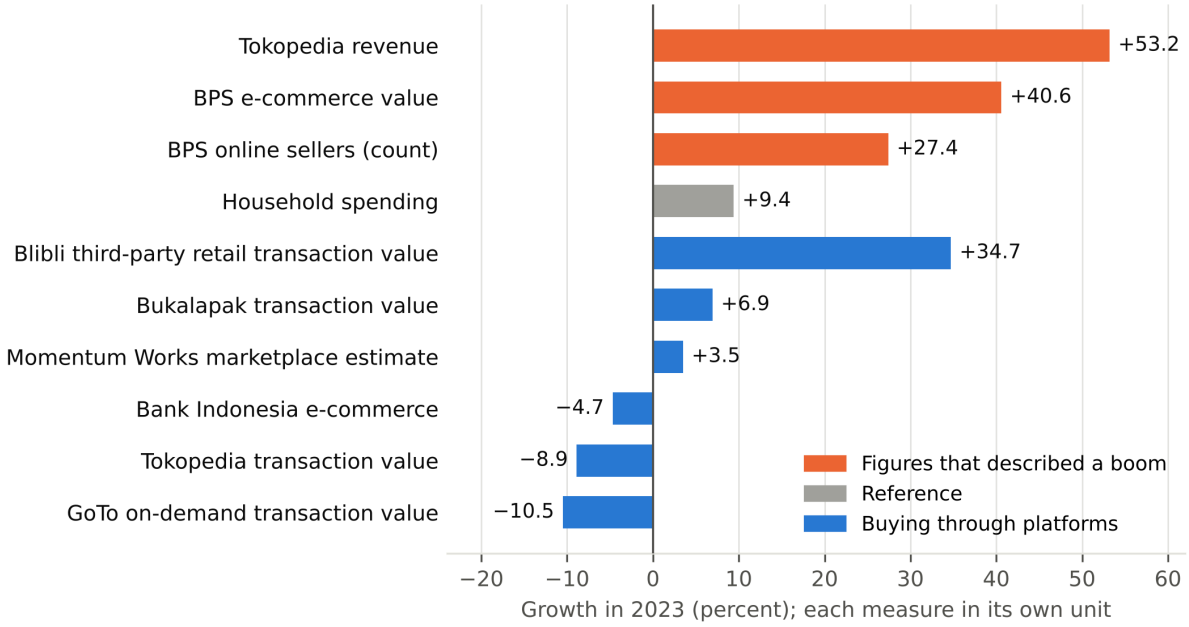



**Figure 5.3. Growth in 2023 (percent).** Blue bars show measures of buying; orange bars show the figures that described a boom. Each measure is in its own unit. Sources: company reports, Bank Indonesia, Momentum Works and BPS; household spending from national accounts. GoTo's change rests on an estimated 2022 value, and Momentum Works' 3.5 is the author's calculation from its published levels (its 2024 release states 3.7). Appendix D gives the values.

Sellers' own accounts of their sales show no boom either, a registered result. BPS asks each business how its online sales changed. For businesses already selling, the registered prediction set an upper limit on the median change. The limit was about 10 percent, the published growth in sales per business. The median change was zero.

Fewer sellers reported a rise than a year earlier. About a quarter reported a rise and about a third a fall. A year before, more than a third had reported a rise and a quarter a fall. The mean change was about −4 percent. Lack of demand became the main obstacle, named by 41 percent of sellers against 35 percent in 2022. The published count of online sellers nonetheless rose by 27 percent in 2023. A rise of that size, in a year without a boom in sales, needs explaining.

## 5.6 H2 and the entry test

H2 holds on BPS's published figures, but the registered entry test shows that first-time entry cannot explain the whole rise in the count. From 2022 to 2023, online sales value grew by 41 percent and the number of businesses by 27 percent. More businesses account for 71 percent of that growth, and higher sales per business for the rest.

H2 holds again from 2023 to 2024. Value grew by 17 percent, and more businesses account for 90 percent of the growth, or 71 percent on BPS's other count for 2023. The split of equation (7) is arithmetic, and it cannot tell a new seller from one counted for the first time.

The registered entry test separates new sellers from sellers counted for the first time, and it fails. In the survey for 2023 (the 2024 file), about 13 percent of sellers began selling online that year. The test required 21.5 percent. Even with no sellers stopping, that many new sellers would raise the count by at most about 16 percent, against a published rise of 27 percent.

Recall error does not explain the shortfall. As Section 3.5 noted, sellers who report a start date later than the true one make new sellers look more numerous. Such errors push the share toward passing, so the shortfall is, if anything, understated. Part of the 2023 rise in the count is something other than first-time entry, and its size is the next result.

## 5.7 What the rise in the count contains

About half of the 2023 rise in the count is not explained by first-time entry, an exploratory result. Table 5.5 sets out the accounting. In the survey for 2023, sellers who began before 2023 outnumber the whole count for 2022 by about 306 thousand. They make up 37 percent of the rise if no seller stopped, and 56 percent if sellers stopped at about 5 percent a year.

**Table 5.5. The 2023 rise in the count of online sellers (thousands)**

| Part of the rise | If no seller stopped | If sellers stopped at 5 percent a year |
|---|---|---|
| Rise between the weighted totals of the two files | 819 (100%) | 819 (100%) |
| Net entry: sellers who began in 2023, less those who stopped | 513 (63%) | 361 (44%) |
| Earlier sellers newly counted above the whole 2022 count | 306 (37%) | 458 (56%) |

*Note:* BPS's published counts give a rise of 821 thousand (BPS 2025a). Source: BPS survey files, weighted; author's calculations.

The cohort counts show the break directly. A cohort should only shrink between surveys, and the cohorts did so in every interval except 2023. Sellers who began by 2020 fell by about 10 percent between the surveys for 2020 and 2022. In the survey for 2023 they rose by 6 percent.

The count behaved normally again in BPS's published table for 2024. The cohort that began before 2020 shrank by about 10 percent. At least 754 thousand new sellers appeared, more than the rise of 584 thousand.

The break was uneven across places and channels. Across the 34 provinces, growth in the count had almost no link to the share of new sellers, a rank correlation of 0.10. It had only a weak link to growth in the sample, at 0.35. A rank correlation compares the order of provinces on two measures and runs from −1 to 1. The break sat in chat and social selling. Among sellers who began by 2022, sellers using only chat or social media rose by 13 percent, and marketplace users by 0.3 percent.

Sampling noise is very unlikely to explain the break. BPS's published error rates put the usual size of the sampling noise in the national count at about 1.4 percent. The 306 thousand extra sellers are about 4.6 times that noise in the change between the two years. The calculation applies the 2024 edition's error rates to both years and treats the two surveys as independent.

Most of the 2023 rise in sales value cannot be traced to what sellers reported. BPS's online sales value rose by 41 percent in rupiah, from about US$53 billion to about US$72 billion at each year's exchange rate. The thesis splits that rise by giving each kind of seller its typical size in the 2022 file. New sellers, at the size of those that began in 2022, account for 25 to 30 percent of the rise. The extra earlier sellers, at the size of chat and social sellers, account for 15 to 17 percent. The typical seller already online reported no growth, with a mean change of about −4 percent when each seller counts once. So 53 to 59 percent of the rise has no source in the sellers' answers. A few large sellers growing fast could fill that part, and the files cannot show whether they did.

The extra count also bears on H2, in an exploratory follow-up. H2's split treats every seller in the count as having entered. Suppose instead that the sellers not explained by entry were reached by wider coverage, and that their sales came with them. Take both out of the 2023 figures. More businesses then account for about 54 percent of the growth if no seller stopped, and about 42 percent if sellers stopped at 5 percent a year. H2 requires more than 50 percent. So H2 holds on the published series, but it sits on its threshold once the extra count is read as coverage. Taking out the sellers but keeping their sales gives 34 to 47 percent, though that version treats the count and the value unevenly.

The checks point to a survey that reached more chat and social sellers in 2023, the kind of break that Section 3.5 described. The files cannot separate that reading from sellers returning after a pause. Who the extra sellers were bears on the choice between the two.

## 5.8 Who the extra sellers were

The extra count leaned toward long-running businesses, older owners and places outside Java, an exploratory result. Each group's extra count compares two numbers (Section 4.7). The first is the group's 2023 count. The second is the 2022 count grown at the group's own yearly rate from 2020 to 2022. The extra count is the first less the second. The extra sellers total about 450 thousand, close to the 458 thousand in Table 5.5. Figure 5.4 compares each group's share of the extra count with its share of the 2022 count.

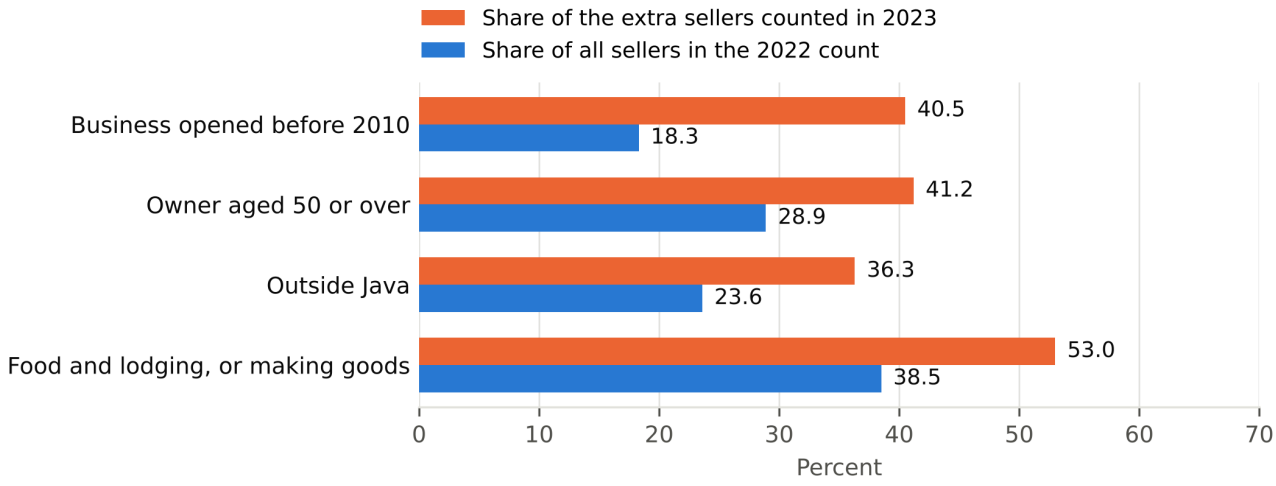



**Figure 5.4. Who the extra sellers were.** The extra sellers are the 2023 count above what each group's normal yearly change in 2020 to 2022 would give, about 450 thousand in total. Source: BPS survey files, weighted; author's calculations. Exploratory.

The extra count did not lean toward young businesses. A lean toward men in the raw counts disappears once each group's normal change is taken out. This profile bears on whether the extra sellers were returning after a pause.

Part-year selling weighs against returners. A returning seller would usually sell for only part of the year, unless it restarted in January, yet part-year selling fell in 2023. Among sellers who began by 2021, those selling for part of the year fell from about 470 thousand to about 320 thousand. The same fall appears for the groups that began by 2020 and by 2022.

The age profile weighs against returners as well. Small firms that stop are more often young firms with young owners (McKenzie and Paffhausen 2019). Returners would then make the extra count lean young. It leaned old.

Returning sellers still cannot be ruled out. Returners equal to 10 to 15 percent of the 2022 count would remove the unexplained part, which is a scenario and not an estimate. The files carry no linked histories, and BPS changed how it asks about months between the rounds. Exits, or a move to full-year selling, could also offset part-year returners. Where online sellers sell, and what records they keep, complete the survey evidence.

## 5.9 Channels and records

Most online selling happens outside marketplaces, and records are thinner there. The path of marketplace use, the value held by sellers without financial statements and the World Bank results for Indonesia were registered. The comparison within the group that began before 2020 is exploratory.

Nearly all online sellers, about 96 percent, use chat or social media. Marketplace use fell every time it was measured, from about 22 percent of sellers in 2020 to about 17 percent in 2024. BPS worded the question slightly differently for 2023 and 2024.

Marketplaces carry only about a fifth of online sales value. The share was 23 to 26 percent in 2022 on the survey files. In BPS's published figures it was 18 percent in 2023 and 16 percent in 2024.

Almost all of the recent growth in value came from outside the marketplaces. From 2023 to 2024, BPS's marketplace value grew by about 1.5 percent and other channels by about 21 percent. BPS's marketplace category is itself broad: 43 percent of its sellers use only Gojek or Grab, and 47 percent use a shopping marketplace.

Records are thinner outside the marketplaces. In 2023, about 29 percent of marketplace sellers kept complete financial statements, against about 12 percent of other sellers. Figure 5.5 shows the share of all sellers who keep them falling. The fall from 2020 to 2022 came while the survey appears to have counted the same kinds of seller. The fall in 2023 may partly reflect the chat-only sellers added to the count.

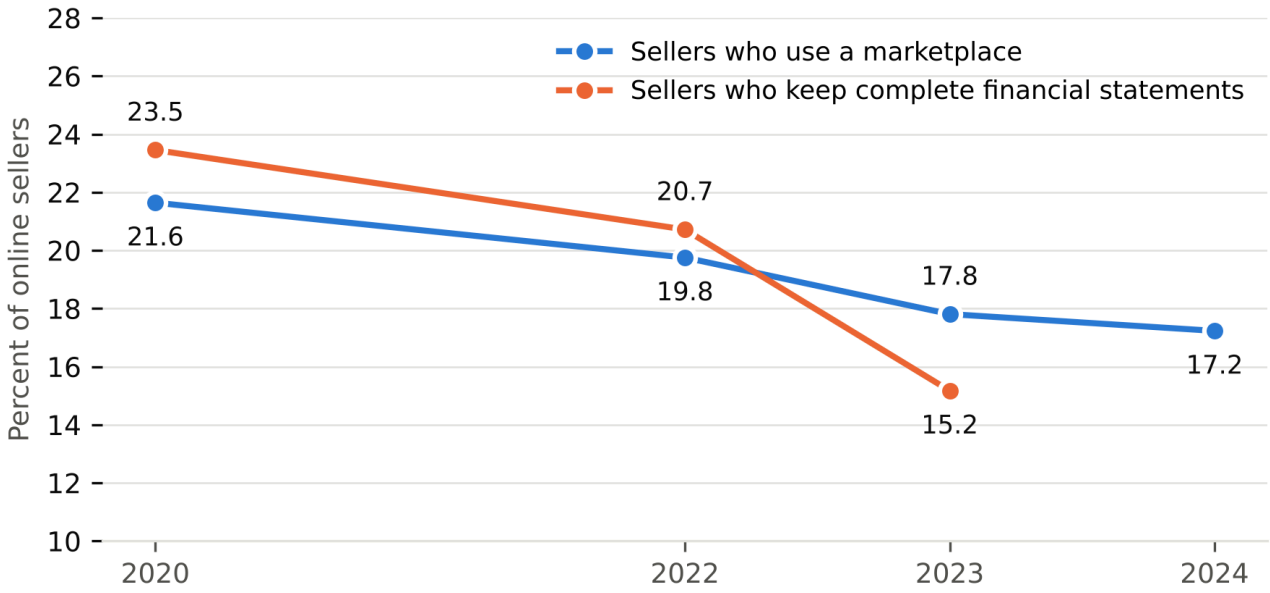



**Figure 5.5. Marketplace use and financial statements, 2020 to 2024.** Marketplace use for 2020, 2022 and 2023 comes from the survey files and for 2024 from BPS's published figure. Financial statements come from the survey files. BPS worded the marketplace question slightly differently for 2023 and 2024, and the survey has no point for 2021. Source: BPS; author's calculations.

The fall reaches long-running sellers too, an exploratory finding. Among sellers who began before 2020, the share keeping complete statements fell from about 23 percent in 2022 to about 17 percent in 2023.

Sellers without statements hold far less of the value than of the count. On the bracket scenarios of Section 4.8, they held 26 to 40 percent of online value in 2022, against 79 percent of the count. The validation rule asks the total built from the brackets to lie within half and double BPS's total. The annual reading passes, just inside the limit at about twice BPS's total, while the monthly reading gives about 23 times. The lower brackets alone exceed BPS's total, so the shares are ranges.

One registered prediction on value failed narrowly. It expected the value share of sellers without statements to fall by no more than 5 points between 2020 and 2022. The share fell by 5.8 points on the main version and by 3.3 points on a second version.

World Bank surveys of Indonesian firms show the same thin records. In six Indonesian cities, 27 percent of unregistered firms use social media with their customers. Among those, 70 percent keep no written records, and their median sales are about US$4,000 a year. A registered prediction that their mean sales would stay below about US$6,600 failed, at about US$7,300.

Among formal firms, about 40 percent in Indonesia have neither a website nor online tax filing. That is the highest share of seven countries, against about 9 percent in Malaysia and in Thailand, a descriptive comparison.

The split by channel bears on the third question of Section 2.4, because a rule built on platform records reaches only the marketplace part. Chapter 6 measures that reach. The same split also marks where the national figures part.

## 5.10 Why the national figures disagree

The national figures agree that growth was small only where they cover the same sellers, and they do not agree on scale. In 2023, Bank Indonesia's e-commerce figure fell by about 5 percent while BPS's rose by about 41 percent. That is the disagreement with which Chapter 1 began, and the third objective asks

what lies behind it.

Where Bank Indonesia's figure and BPS's cover the same ground, the growth is small. From 2022 to 2023, BPS's own marketplace slice grew by about 0.2 percent in the middle scenario. Across the scenarios it grew by 0.2 to 13 percent. From 2023 to 2024 it grew by about 1.5 percent. Most of the disagreement sits outside the marketplaces, among the sellers that only BPS reaches.

A registered test on size gives a mixed answer. It predicted that the part of BPS's marketplace sales that went to final consumers in 2022 would fall below Bank Indonesia's figure. The prediction holds in the low and middle scenarios, at 0.16 and 0.68 times Bank Indonesia's figure. It fails in the high one, at 1.62 times.

In 2024 the figures divide by coverage. Those that cover mainly platforms grew by 5 to 7 percent, and BPS's grew by about 17 percent. The adjacent figures of Table 4.3 point the same way as BPS but do not measure Indonesian e-commerce. Parcels shipped in Southeast Asia grew by about 25 percent, and the tax on foreign digital services by about 25 percent.

Payment figures grew faster than any of these from 2023 to 2024. E-money shopping grew by about 30 percent, mobile-banking purchases by 83 percent and QRIS payments by 187 percent. They are context only, because they include in-person payments and the switch away from cash.

Levels cannot be reconciled. In 2024, BPS put online sales at about US$81 billion and Bank Indonesia at about US$31 billion. In 2023, BPS's marketplace slice was about US$13 billion, against about US$38 billion for Tokopedia and the Shopee estimate together. In 2024 it was less than a quarter of Momentum Works' marketplace estimate of about US$57 billion.

Platform figures and the seller survey count different things. Platform transaction value includes shipping, digital goods, travel and offline sales that a seller survey does not count in the same way. The counts of sellers differ too. Tokopedia alone said that it had more than 14 million merchants in 2023, while BPS counted about 0.7 million marketplace sellers that year (GoTo Gojek Tokopedia 2024b).

Platform commerce does not always keep pace with household spending. In each of the four quarters to mid-2026, GoTo's on-demand transaction value grew more slowly than household spending. In the last, it grew by 2 percent against 8 percent, while its revenue grew by about 20 percent. Grab's and Shopee's figures grew faster, but they cover the region.

The results answer the research question at two levels. On the inner link, revenue showed a small part of the commerce and stopped being a fair guide when the take rate moved. On the outer link, the 2023 rise in the count of sellers was only partly first-time entry. The figures part most where only the survey looks. Chapter 6 turns to the rules that Indonesia is writing.

# Chapter 6. The Rules and How Far They Reach

Indonesia's government is writing rules that act on the figures this thesis measures. In 2026 it capped a ride-hailing commission and limited discounts in online trade. Since October 2026, four marketplaces have collected a tax from their sellers. Platforms must also send data to BPS, and BPS has run a new economic census. Each rule acts on one part of what Chapter 5 measured, and each reaches only that part.

The marketplace tax is the rule aimed most directly at online sellers. It can collect from at most about 1 in 20 online sellers and about a fifth of online sales value. It also gives the state a record of every seller on the four marketplaces, including those who pay nothing. This chapter measures what each rule can reach, using the data of Chapter 5. What each rule changes can be seen only after it takes effect, and Section 6.5 sets out the predictions that will be scored on those results.

## 6.1 The rules and what each acts on

The recent rules act on three parts of the figures: the platforms' take rate, the sales of sellers inside marketplaces, and BPS's coverage. Table 6.1 lists each rule with the part it acts on and the section of Chapter 5 that measures that part.

**Table 6.1. Indonesia's recent rules and what each acts on**

| Rule | What it does | What it acts on | Measured in | Status of the text |
|---|---|---|---|---|
| Commission cap (Transport Minister Decree KP-PHB 532/2026; Presidential Regulation 27/2026) | Caps the commission on two-wheel passenger trips at 8 percent from 1 July 2026 | The fee rate, for one service | 5.3 | Decree in force from July 2026; presidential regulation released 8 October 2026; neither text public when this thesis was written |
| Discount rule (Trade Minister Regulation 19/2026, Art. 18) | Requires every online trade operator, including ride-hailing apps that sell goods, to prevent repeated price subsidies that hurt local products and endless below-cost discounts | The incentive rate | 5.3 | Primary text read |
| Marketplace tax (Finance Minister Regulation 37/2025) | Designated marketplaces collect 0.5 percent of domestic sellers' gross sales; individuals with sales up to Rp500 million a year are exempt; every seller gives a tax number or national ID | Sellers' sales inside marketplaces | 5.9 | Primary text read; collected by Shopee, Tokopedia, Lazada and Blibli since 1 October 2026 |
| Small-business final tax (Government Regulation 20/2026) | Keeps a 0.5 percent final tax on turnover up to Rp4.8 billion a year; excludes content creators | Small sellers in every channel | Not measured | Primary text read |
| Platform data to BPS (BPS Regulation 4/2023) | E-commerce operators must send BPS their revenue, transactions and numbers of sellers and buyers, with sanctions | BPS's coverage | 5.10 | Primary text read |
| Economic census 2026 | Door-to-door count of all businesses, including online selling, affiliates and influencers | BPS's sampling frame | 5.7 | News reports; results not yet released |
| Ban on payments on social media (Trade Minister Regulation 31/2023, carried into Regulation 19/2026) | Bars social media platforms from handling payments for sales | Where online sales are paid | 5.9 | Primary text read |

*Note:* At the 2024 rate, Rp500 million is about US$32,000 and Rp4.8 billion about US$300,000. Section 4.8 converts the exemption at the 2022 rate, about US$34,000, to match the survey year. Source: the legal texts; news reports where marked.

Equation (6) shows where the first two rules act. The take rate is the fee rate less the incentive rate. A cap on a fee pushes the take rate down. A limit on incentives pushes it up. The two rules act on opposite sides of the same take rate.

The marketplace tax acts further out, on the sales of sellers inside marketplaces. It leaves the take rate alone, but it depends on the marketplaces' records of each sale. The platform-data rule and the census act on the national figure: the first on what BPS receives, the second on where BPS looks.

Two rules serve as context. The first is the 2023 ban on payments on social media, the step with which Chapter 1 began. Selling through chat and social media remained the norm, and in 2023 about 96 percent of online sellers used them. The second is the small-business tax, which applies in every channel. The survey files cannot identify the content creators it excludes, so the thesis does not measure its reach.

The sections that follow take the other rules in the order of Figure 3.1, from the take rate outward.

## 6.2 What the commission cap and the discount rule can move

The commission cap reaches a small part of the take rates this thesis measures. It followed protests by ride-hailing drivers, who since 2025 had asked that platforms keep at most 10 percent of each fare, down from 20 (Ronal 2025). From July 2026, a platform may keep at most 8 percent of the fare on a two-wheel ride. The cap does not cover car rides, food delivery or marketplaces. The presidential regulation released in October 2026 asks another ministry to set delivery fees by the end of the year.

The trips under the cap are a small share of each company's business. At GoTo, the two-wheel ride service brings about 7 percent of group net revenue (GoTo Gojek Tokopedia 2026). At Grab, two-wheel rides in Indonesia are about 6 percent of the transaction value of its rides business across the region (Grab Holdings Limited 2026). That is about 2 percent of Grab's on-demand transaction value. Both companies report rides and deliveries as one on-demand take rate, so the cap acts on a slice of the figure Chapter 5 measured.

The arithmetic of the cap shows how small that slice is. At Grab, the cut from 20 to 8 percent applies to trips worth about 2 percent of on-demand transaction value. If nothing else changes, it lowers the on-demand take rate by at most about a quarter of a point. Grab's on-demand take rate is about 13 percent. At GoTo, where two-wheel rides weigh more, the same arithmetic gives about 1 to 1.4 points on a take rate of about 22 percent. In both cases the wedge grows by the same amount. The cap moves money from the platform's revenue to the drivers' fares, inside the same transaction value.

The players around the cap have different stakes. Drivers asked for it, and the President announced the limit at a May Day rally in 2026, saying the cut "must be below 10 percent" (CNN Indonesia 2026b). The platforms lose part of the commission unless they recover it elsewhere, for example by cutting incentives. The cap brings the state no revenue; its tax on online sellers works through the marketplaces instead (Section 6.3).

A cap on one fee does not fix the take rate, because the platform still sets its incentives. At GoTo, on-demand revenue rose by about 20 percent in the second quarter of 2026 while transaction value grew by 2 percent. The company attributed the rise to its mix of services and to "rationalization of incentives as we prepared for the implementation of the 8% commission rate" (GoTo Gojek Tokopedia 2026). Its take rate rose to about 22 percent in that quarter, the last before the cap, from about 18 percent a year earlier.

The discount rule acts on the lever that moved take rates most often. Lower incentives were the larger part of the rise in the take rate in four of the seven windows of Table 5.4. The rule binds every online trade operator, including ride-hailing apps that sell food or goods. Whether it reaches ride fares, which the transport ministry sets, is unclear.

Where incentives can be followed over time, much of that lever had been pulled before the rule came. Tokopedia, the only marketplace that reported its incentives, cut them by about a third in 2023, its last full year of separate reporting. At GoTo's on-demand business, incentives fell from about 11 percent of transaction value in 2023 to about 4 percent in 2025. Grab still spent about 10 percent of on-demand transaction value on incentives in the four quarters to early 2026, across the region. The thesis did not register a test of the discount rule, because its effect on the take rate cannot be told apart from changes in fees.

Marketplace fees moved the other way. Shopee, Tokopedia and TikTok Shop added a fixed fee to every order in mid-2025 (Islamiati 2025). In 2026, Shopee and TikTok Shop, which took over Tokopedia in 2024, raised their seller fees again (CNN Indonesia 2026a; Setyowati 2026). The government declined to cap these fees. It later required marketplaces to halve the service fee for small sellers of local products

who apply for the discount (Ayudiana 2026). The rules on the take rate cap one fee on one ride-hailing service and leave the largest marketplace fees free. What the companies' take rates did after the cap is the subject of the predictions in Section 6.5.

## 6.3 How far the marketplace tax reaches

The marketplace tax can collect from few online sellers, but it gives the state a record of every seller on the four marketplaces that collect it. These marketplaces collect 0.5 percent of each domestic seller's gross sales and pass it to the tax office. Individual sellers with sales up to about US$34,000 a year are exempt (Section 4.8). Collection started in October 2026 through Shopee, Tokopedia, Lazada and Blibli (Directorate General of Taxes 2026c). It had begun once in August 2026, then stopped, and the tax collected was returned (Directorate General of Taxes 2026a, 2026b).

At most about 1 in 20 online sellers falls under the tax. In 2022, marketplace sellers with sales above BPS's bracket line of about US$20,000 a year were about 5 percent of all online sellers. The exemption sits higher, at about US$34,000, so the true share is smaller.

The value the tax reaches is larger, but it is still a minority of online sales. Marketplaces carried 23 to 26 percent of online sales value in 2022, and sellers above the bracket line held 77 to 94 percent of that. So the tax reaches at most about 18 to 24 percent of online sales value, about a fifth. Both figures use BPS's marketplace category, which also counts food and ride apps. Those apps are not among the four collectors, so both figures are upper limits for a second reason.

The record reaches further than the tax. Every domestic seller on a designated marketplace must give the marketplace a tax number or national ID and an address before being paid. Exempt sellers file a statement as well. Tax research finds that records held by a third party improve compliance where they exist (Section 2.3).

The record still covers a minority of sellers. Marketplace sellers were about 18 percent of online sellers in 2023. Only 47 percent of them used a shopping marketplace, and most of the rest sold only through Gojek or Grab. So the record covers at most about 8 percent of online sellers, including the many who pay nothing.

The tax leaves most sellers where they were. Sellers off the marketplaces sell through chat, social media or their own websites, and they keep complete financial statements less often (Section 5.9). Few of them plan to move. In the survey for 2023, about 13 percent of sellers off the marketplaces wanted to join one.

The revenue at stake is small. At 0.5 percent, the tax would raise about US$60 million a year on BPS's marketplace sales for 2024, before exemptions. On e-Conomy's estimate of e-commerce transaction value, about five times larger, it would raise about US$300 million. Even the larger amount is about 0.2 percent of the government's tax target for 2026.

The tax adds a record more than revenue. Whether sellers move away from the four marketplaces now that collection has started will be known only from later data.

## 6.4 What the statistics rules can change

The rules on statistics act on the outer link, where the national figures parted in 2023. Since 2023, e-commerce operators must send BPS their revenue, transactions and numbers of sellers and buyers. Operators that do not comply face sanctions. By late 2024, 61 operators had sent data and 75 registered operators had not (Risalah 2024). BPS's publication for 2024 describes its figures as survey-based and does not mention the platform data.

Full compliance would still leave most online selling to the survey. Platform data describe the marketplace slice, which holds about a fifth of online sales value and under a fifth of sellers (Section 5.9). The chat and social sellers, where the national figures part (Section 5.10), would still be counted only by BPS's survey.

The economic census of 2026 acts on where the survey looks. BPS's e-commerce survey draws its areas from the economic census of 2016. The 2026 census visited homes and businesses from June to August 2026. It asked about online selling and counted affiliates and influencers. Its results had not been released when this thesis was written.

A new frame can move the count of sellers without any change in selling. About half of the 2023 rise was not explained by first-time entry (Section 5.7). When BPS adopts a frame from the 2026 census, counts on either side of the change will need the same care. The entry test of Section 3.5 can be repeated on the first survey that uses the new frame.

## 6.5 What the registered predictions test

Nine predictions on the rules and on what they act on were registered before any result for the third quarter of 2026 existed. Eight concern the third-quarter results of GoTo, Grab and Sea. Six of them bear on the commission cap, one on marketplace fees and one on buying through platforms. The ninth concerns BPS's next figure for the marketplace share. Table 6.2 lists them with the date each is scored.

**Table 6.2. Registered predictions on the 2026 rules**

| Prediction | What it expects | Bears on | Scored | Outcome |
|---|---|---|---|---|
| P1 | GoTo's on-demand take rate falls by 0.3 to 2.5 points from the second quarter of 2026 | Commission cap | Late October 2026 | [to be filled] |
| P3 | GoTo's on-demand transaction value grows by 0 to 12 percent on a year earlier | Commission cap | Late October 2026 | [to be filled] |
| P5 | GoTo's on-demand transaction value grows by at least 5 percent; the prediction fails only below 2 percent | Commission cap, with spending on growth | Late October 2026 | [to be filled] |
| R1 | GoTo's on-demand revenue growth less transaction value growth lies between −0.5 and +12.5 points; it follows from P1 and P3 | Commission cap | Late October 2026 | [to be filled] |
| R2 | GoTo's on-demand transaction value grows more slowly than household spending | Buying through platforms | Early November 2026 | [to be filled] |
| P2 | Grab's on-demand take rate stays within −0.6 to +0.4 points of the second quarter | Commission cap | Mid-November 2026 | [to be filled] |
| P4 | Grab's change in the rides take rate, less its change in the deliveries take rate, is at most +0.3 points | Commission cap | Mid-November 2026 | [to be filled] |
| R3 | Shopee's revenue growth less its transaction value growth is at least +10 points | Marketplace fees | Mid-November 2026 | [to be filled] |
| Marketplace share | BPS's next marketplace share of online sales value is at most 15.79 percent | Reach of marketplace rules | Next BPS release | [to be filled] |

*Note:* Ranges as registered; registration dates are in Appendix C. A tenth prediction, that the start of the marketplace tax would slip again, is withdrawn: collection had begun on 1 October 2026, a day before the prediction was registered. Sources for scoring: company results releases; BPS national accounts and e-commerce publication.

The predictions describe company numbers, not the effect of the cap. Both companies report rides and deliveries together, so any effect of the cap is diluted in their figures. P1 and P5 are read together. If both hold, GoTo met the cap partly by spending more to grow. If P1 holds and P5 fails, GoTo let the lower commission show in its take rate without spending more to grow. P4 has limited power, because its range cannot separate the cap from ordinary quarter-to-quarter noise. R2 and R3 concern single companies, and Shopee's figures cover Asia and Brazil.

> **[Results slot]** To be written after the GoTo results (late October), the BPS national accounts (early November) and the Grab and Sea results (mid-November). For each outcome: the result against its range in Table 6.2 and Appendix C, then one paragraph on what the outcomes show about the cap and one on marketplace fees.

The outcomes will be reported whatever their direction, as the other registered tests are.

## 6.6 Beyond Indonesia

Other countries are building rules on platform records too, and the same question of reach applies to them. The OECD's model rules ask platforms to report what their sellers earn, and a later module added the sale of goods (OECD 2020, 2021). The European Union applies such rules through its DAC7 directive, with reports on activity from 2023 (European Union 2021). The Philippines has marketplaces and payment apps withhold tax from sellers' payouts, with small sellers exempt (Bureau of Internal Revenue 2023). Viet Nam has had platforms with a payment function withhold tax for household sellers since 2025 (Government of Viet Nam 2025).

Indonesia's design goes further than reporting, because its marketplaces collect the tax as well as identify the seller. Every one of these rules reaches only the sales that pass through a platform. Where most selling runs through platforms, a rule built on their records reaches most sellers. Where it runs through chat and social media, as in Indonesia, it reaches a minority.

Commission caps have been tried elsewhere as well. Fourteen US cities and states capped the commissions that delivery platforms charged independent restaurants. The platforms raised delivery fees for consumers and showed chain restaurants first more often. Independent restaurants lost orders and revenue, while chains gained (Li and Wang 2025). In the terms of equation (6), a cap on one fee left the platform free to change another. Indonesia's cap also covers one fee on one service.

The thesis does not measure the split between channels outside Indonesia. Any statistics office that asks sellers which channels they use can make the same count, and with it measure how far its own rules reach.

# Chapter 7. Conclusion

Each figure of Indonesia's online economy is a fair guide to another only while a condition holds. Revenue follows transaction value while the take rate stays steady. A survey's count of sellers follows entry while the survey's reach stays steady. Between 2022 and 2024 the platforms' take rates moved, and revenue parted from the commerce it carried. In 2023 about half of the rise in BPS's count of online sellers was not explained by first-time entry.

This chapter answers the research question at both links of Figure 3.1. It then says what the answer adds for those who use the figures, reports the predictions that failed, states the limits of the evidence and sets out the next steps.

## 7.1 The answer to the research question

The research question asked how much economic activity remains invisible when digital platforms are measured through their reported revenue, and when that choice changes the conclusion about growth. The answer to the first half is that most of the activity remains outside revenue. In 2023, three major platforms in Indonesia kept about US$3 billion of the US$43 billion that passed through them. The invisible wedge was about US$40 billion, and it moved almost one for one with the commerce.

Platform commerce is itself only part of online commerce. In BPS's survey, marketplaces carry about a fifth of online sales value. Any measure built on platforms, whether revenue or transaction value, leaves out most online sales, which move through chat, social media and sellers' own websites.

The answer to the second half differs by link. On the inner link, the choice of revenue changed the conclusion between 2022 and 2024. The Indonesian and regional platforms changed their take rates about three times as much as platforms abroad, measured against the take rate itself, so H1 is rejected. They keep a much thinner share of each sale, so price changes of a similar size in points moved their revenue far more. In the windows traced, pricing moved revenue more than commerce did.

On the outer link, H2 holds on BPS's published figures: more businesses account for most of the growth in online sales value. The registered entry test failed, and about half of the 2023 rise in the count is not explained by first-time entry. The extra count leaned toward long-running businesses. Read as wider coverage, it leaves H2 on its threshold. With those sellers and their sales taken out, more businesses account for about 42 to 54 percent of the growth, against the 50 percent H2 requires. In the same year, five of six measures of buying through platforms grew more slowly than household spending, and the median seller already online reported no change in sales. Reading BPS's 2023 figures as a boom in online selling overstates what sellers and platforms reported.

The national figures agree that growth was small where they cover the same sellers. They part outside the marketplaces, among the chat and social sellers that only BPS reaches. The figure a reader picks decides whether 2023 looks like a stall or a boom. Market value, for its part, moved with transaction value, and the data do not show that markets valued growth from pricing any less.

## 7.2 What the thesis adds

For investors, the thesis adds a check that can be run on any platform that reports both amounts. Revenue growth splits into more commerce and a larger take rate with two public figures (equation (5)). In Indonesia, the take rate accounted for about two thirds of a revenue move at the typical pair, against about a quarter abroad. The thinner the take rate, the more a small change in fees or incentives moves revenue growth. The same split shows when market value per dollar of revenue and per dollar

of commerce will disagree. Across 28 listed platforms, market value moved with transaction value, as a registered prediction stated. A 1 percent rise in transaction value went with about 0.6 percent more market value. Earlier studies used web traffic as the measure of activity; here it is the transaction value that platforms report themselves.

For statistics offices, the thesis adds a test of what a count of sellers represents. The entry test needs no linked records, only a question on when each business began selling online. Applied to BPS's survey, it showed that the 2023 rise in the count was only partly first-time entry. The same test can mark a break in any survey that asks the question.

For tax authorities and regulators, the thesis measures how far rules built on platform records reach. In Indonesia, the marketplace tax can collect from at most about 1 in 20 online sellers and about a fifth of online sales value. It also gives the state a record of every seller on the four marketplaces that collect it, at most about 8 percent of online sellers. Each of the other recent rules acts on one part of the take rate or of the national figure, and Chapter 6 maps each to the part it reaches.

For research, the thesis adds connected evidence for one country: the platforms' accounts, the sellers' own answers and the rules that act on both. It tracks the invisible wedge and growth divergence over time, and its ecosystem ratio is the author's own measure. All three use only figures that platforms already publish, so they can be computed for any platform that reports transaction value.

## 7.3 Predictions that failed

Several registered predictions failed, and the failures are part of the answer. The entry test is the clearest case. It required at least 21.5 percent of the sellers counted for 2023 to have begun that year, and the survey gave about 13 percent. That failure is the main result of the outer link.

The test of H1 with quarterly pairs did not reject H1 ($p = 0.19$). The quarterly series cover mostly 2023 to 2026, after most of the repricing. So the rejection of H1 describes a period, 2022 to 2024, and not a lasting trait of Indonesian platforms.

The second market-value prediction failed in the main sample ($p = 0.20$). It expected markets to value growth from pricing less than growth from more commerce. It held only in a second run that mixes segment figures with whole-company market values. Whether investors price the two sources of growth differently remains open.

Three predictions on sellers' records and sales failed narrowly or in part. The value share of sellers without financial statements fell by 5.8 points between 2020 and 2022, beyond the registered limit of 5. On a second version it fell by only 3.3 points. The mean sales of unregistered firms that use social media were about US$7,300 a year, above the registered limit of about US$6,600, while their median was about US$4,000. The comparison with Bank Indonesia's figure held in two of three value scenarios and failed in the third.

A registered rule on discounts gave only thin support. It expected large take-rate moves only where a platform's incentives were at least as large as its revenue. Out of sample, the rule pointed the right way, but it rested on two cases and failed at its main threshold.

Appendix C lists every registered test with its outcome. The predictions on the 2026 rules are scored in Section 6.5. One of them, that the start of the marketplace tax would slip again, was withdrawn, because the tax had started a day before the prediction was registered. A comparison of seven countries on firms' digital records was withdrawn as a test for a similar reason. Published figures seen at registration already settled it, so it is reported as descriptive.

## 7.4 Limits of the evidence

The invisible wedge is the part of each sale that a platform passes on to sellers, drivers and others. It is not missing output, unrecorded income or unpaid tax. The thesis does not find BPS's figures wrong in general, nor that total online buying in Indonesia fell. It finds that one year's count of sellers moved for reasons other than first-time entry, and that the figures part where they cover different sellers. The splits of Table 5.4 are accounting, not causes. Marketplace sellers keep financial statements more often, but the thesis does not show that marketplaces lead them to.

The platform evidence rests on a small sample. Three Indonesian marketplaces report both revenue and transaction value. The main test of H1 adds the on-demand businesses of Grab and GoTo, which mix rides and deliveries, and Grab's figures cover the whole region. That test was set up after a narrower first test, so it is exploratory. The tests of H1 are one-sided, because each asks whether the Indonesian and regional group moves more. A two-sided test would about double each p-value, and the main Mann–Whitney result would still be below 1 percent.

The platform figures carry their own limits. Shopee's and Grab's Indonesian amounts rest on an outside estimate, an assumed take rate or a company-wide ratio, and they stay out of the main tests. Grab revised some figures between filings. The thesis takes each year from the latest filing that reports it, so one series can mix filings from before and after a revision. Blibli's incentives are a proxy, and Bukalapak's reports do not allow the parts of a revenue move to be separated. Company explanations of why revenue moved are reported as statements and are not tested. Foreign platforms also differ in how they book shopper discounts. Eight of the 27 record them mainly as marketing costs, not as lower revenue, which could steady their take rates. Their take rates were not steadier, and without them the test of H1 still gives $p = 0.004$, an exploratory check (Appendix B).

The survey files describe sellers in one year, not businesses followed over time. No business identifier links the files, so the thesis cannot follow a seller from one round to the next, and returning sellers cannot be ruled out. Revenue appears only in brackets, and only in the 2021 and 2023 files, which describe 2020 and 2022, so value shares are ranges. The 2024 file, which describes 2023, has no revenue amounts. At the top, sellers with sales above about US$3.4 million a year rest on 19 sample rows in the survey for 2022. Yet they hold 18 to 39 percent of the marketplace value.

Two changes in the questions also limit the survey checks. BPS changed how it asks about the months of selling between the rounds, which affects the check on part-year sellers. The education codes of the survey for 2023 were matched to the earlier groups by their shares, so education there is an inferred grouping.

The accounting of the 2023 rise is exploratory, because it was designed after the files were opened. Its range depends on how many sellers stopped, and it assumes that no seller returned after a pause. Returners equal to 10 to 15 percent of the 2022 count would remove the unexplained part. The check on sampling noise applies one year's published error rates to both years and treats the two surveys as independent. The files carry no sampling-area or design variables, and the thesis did not contact BPS. So a change in BPS's frame cannot be told apart from other changes in coverage, and the break covers a single year.

The national figures are read as published. Bank Indonesia does not publish its method. Momentum Works and e-Conomy publish estimates, not counts. Payment figures include payments made in person. The World Bank's survey of unregistered firms covers six cities, not the whole country.

The market-value result is an association across 28 listed platforms. Market value also changes when a company issues shares, so the result is not a stock return. The thesis does not claim that transaction value causes market value to move.

The rules are mapped, not evaluated. The legal texts behind the commission cap were not public when the thesis was written, so its terms come from official statements, news reports and GoTo's earnings call. The reach of the marketplace tax is an upper limit, because BPS's bracket line sits below the exemption and BPS's marketplace category also counts food and ride apps. The thesis measures what each rule can reach and makes no claim about its effects.

Three of the sources are licensed: the BPS survey files, the LSEG market values and the World Bank survey files. The repository holds the code and the national totals, and anyone who obtains the same files can rerun the analysis. The evidence concerns Indonesia, with regional series labelled as such.

## 7.5 Next steps

The first step is to score the registered predictions on the 2026 results, as Section 6.5 sets out. The outcomes will show what the companies' take rates did after the commission cap. Later data will show whether sellers moved away from the four marketplaces that collect the tax.

Reconciling the levels of BPS's and the platforms' figures is the second step. It needs common definitions of what counts as a sale, including shipping, digital goods, travel and offline sales. The platform data that BPS now collects could supply the platform side, if BPS publishes them.

The third step concerns the 2026 economic census. When BPS adopts a frame from it, the entry test can be repeated on the first survey that uses the new frame. A business identifier that links survey rounds would settle the question of returning sellers.

The fourth step is a stock-return extension. A test of share prices around results announcements would ask whether investors react differently to growth from commerce and growth from pricing. It needs announcement timing and raw prices that can be reproduced.

The platform sample can grow as well. Lazada and TikTok Shop can join the test of H1 once they report Indonesian revenue and transaction value.

The last step takes the channel count abroad. Any country whose statistics office asks sellers which channels they use can measure how much online selling its platform rules reach.

The thesis began with figures that moved apart in 2023. BPS's online sales rose by about 41 percent while Bank Indonesia's measure fell by about 5 percent, and Tokopedia's revenue rose while its transaction value fell. The figures disagreed because they measure different things. Revenue is what a platform keeps. Bank Indonesia's figure comes from large e-commerce firms, and BPS's survey counts the sellers it reaches. Each stops standing for another once a take rate or a survey's reach changes. A reader who checks which condition holds can use any of them.

# Appendices

These appendices hold the detail that the main text points to. Every number comes from a table in the project's repository, named in the notes in small type. Nothing here changes a result in Chapter 5.

## Appendix A. Platforms, windows and the wedge by year

### A.1 The foreign group (27 platforms)

Pairs are consecutive years that pass the rules of Chapter 4, section 4.1: for the platforms added, the later year is not one in which own goods exceed half of revenue; for the eight chosen first, the reporting scope did not change. Ozon's one pair starts in a year that fails the own-goods rule (Appendix B). The median is the platform's median size of yearly change in log take rate. Source: `tables/spinoff_panel.csv`, `tables/transitions_master.csv`.

| Platform | Chosen first or added | Pairs | Median size of yearly change in log take rate |
|---|---|---|---|
| ACV Auctions | Added | 5 | 0.15 |
| Waitr | Added | 6 | 0.10 |
| Alibaba | Added | 7 | 0.21 |
| Despegar | Added | 9 | 0.07 |
| 1stDibs | Added | 6 | 0.03 |
| Expedia | Added | 11 | 0.03 |
| Fiverr | Added | 7 | 0.05 |
| Grubhub | Added | 6 | 0.09 |
| Hepsiburada | Added | 5 | 0.03 |
| Kaspi.kz | Added | 4 | 0.08 |
| Mogu | Added | 9 | 0.21 |
| Ozon | Added | 1 | 0.18 |
| PDD Holdings | Added | 4 | 0.13 |
| RB Global | Added | 11 | 0.06 |
| The RealReal | Added | 8 | 0.04 |
| Vivid Seats | Added | 6 | 0.07 |
| Upwork | Added | 8 | 0.04 |
| Vacasa | Added | 5 | 0.03 |
| Yatra | Added | 11 | 0.03 |
| Etsy | Chosen first | 3 | 0.07 |
| Jumia | Chosen first | 3 | 0.16 |
| Mercado Libre | Chosen first | 2 | 0.06 |
| Rakuten | Chosen first | 1 | 0.02 |
| Sea / Shopee | Chosen first | 7 | 0.25 |

| Platform | Chosen first or added | Pairs | Median size of yearly change in log take rate |
|---|---|---|---|
| Shopify | Chosen first | 5 | 0.05 |
| Zalando | Chosen first | 4 | 0.02 |
| eBay | Chosen first | 4 | 0.03 |

Left out after examination: Dingdong, JD.com, Jumei, Liquidity Services, Newegg and Vipshop, because their own goods are most of revenue in every year; and four platforms that disclose no transaction value (tickers NBIS, TCOM, TOUR and UXIN).

### A.2 The Indonesian and regional group

| Platform | Pairs | Median size of yearly change in log take rate | Share of pairs with a change of 0.10 or less |
|---|---|---|---|
| Blibli third-party retail | 4 | 0.25 | 0.00 |
| Bukalapak Group | 3 | 0.14 | 0.33 |
| GoTo on-demand | 3 | 0.09 | 0.67 |
| Grab on-demand | 4 | 0.20 | 0.50 |
| Tokopedia | 1 | 0.52 | 0.00 |

### A.3 The seven windows traced (Table 5.4 in full)

Shares are of the log growth of revenue. Source: `tables/growth_decomposition.csv`, built from `tables/l_panel.csv` by `growth_decomposition.py`. Tokopedia, GoTo: audited statements. Grab: annual filings, summed over mobility and deliveries. Blibli: incentives are a proxy (gross profit before discounts less net revenue) and stay out of the main sample. GoTo's 2022 transaction value is an estimate (2022 first and second quarters backed out of 2023 year-on-year comparisons).

| Platform and window | Growth of revenue (%) | Growth of transaction value (%) | Take rate, start (%) | Take rate, end (%) | More commerce | Higher fees | Lower incentives |
|---|---|---|---|---|---|---|---|
| Grab on-demand, 2021–24 | 320.5 | 62.3 | 5.34 | 13.83 | 33.7 | 33.8 | 32.5 |
| Grab on-demand, 2022–24 | 85.8 | 30.1 | 9.69 | 13.83 | 42.5 | 10.6 | 46.9 |
| GoTo on-demand, 2023–25 | 114.2 | 22.4 | 10.62 | 18.58 | 26.6 | 6.1 | 67.3 |
| GoTo on-demand, 2022–25 | 110.7 | 9.6 | 9.66 | 18.58 | 12.3 | 36.1 | 51.7 |
| Tokopedia, 2022–23 | 53.2 | -8.9 | 1.48 | 2.48 | -21.9 | 76.5 | 45.4 |
| Blibli third-party retail, 2022–25 | 679.4 | 43.6 | 0.54 | 2.92 | 17.6 | 62.6 | 19.8 |
| Blibli third-party retail, 2022–23 | 465.3 | 34.7 | 0.54 | 2.25 | 17.2 | 20.1 | 62.7 |

Tokopedia, 2022–23, line by line (Rp million, `tables/tokopedia_segment.csv`): gross revenue 8,143,239 to 8,988,909; incentives 4,112,320 to 2,813,719; net revenue 4,030,919 to 6,175,190. Net revenue rose by 2,144,271, of which lower incentives account for 60.6 percent and higher gross revenue for 39.4 percent (about 60 and 40 percent in the text).

### A.4 Wedge and ecosystem ratio by platform-year, Indonesian series

Original currency and units, as in the filings: Rp million for Tokopedia and Bukalapak, Rp billion for Blibli, and US$ billion for Grab and Shopee, whose Indonesian amounts are derived or estimated. Shopee's revenue assumes a take rate of 8.4 percent in 2022, 10 percent in 2023 and 10.8 percent in 2024. Table 5.2 converts Tokopedia's 2023 rupiah amounts to US dollars. Take rate and ecosystem ratio carry no currency. Source: `tables/take_rate_levels.csv`.

| Series | Year | Transaction value | Revenue | Wedge | Take rate (%) | Ecosystem ratio | Label |
|---|---|---|---|---|---|---|---|
| Grab | 2021 | 1.88 | 0.08 | 1.80 | 4.20 | 22.8 | Derived value |
| Grab | 2022 | 3.83 | 0.28 | 3.55 | 7.19 | 12.9 | Derived value |
| Grab | 2023 | 5.38 | 0.60 | 4.78 | 11.24 | 7.9 | Derived value |
| Shopee | 2022 | 18.68 | 1.57 | 17.11 | 8.40 | 10.9 | Derived value |
| Shopee | 2023 | 21.52 | 2.15 | 19.37 | 10.00 | 9.0 | Derived value |
| Shopee | 2024 | 25.99 | 2.81 | 23.18 | 10.80 | 8.3 | Derived value |
| Tokopedia | 2022 | 273,146,166 | 4,030,919 | 269,115,247 | 1.48 | 66.8 | Direct segment disclosure |
| Tokopedia | 2023 | 248,836,156 | 6,175,190 | 242,660,966 | 2.48 | 39.3 | Direct segment disclosure |
| Blibli third-party retail | 2021 | 15,746 | 122.00 | 15,624 | 0.77 | 128.1 | Direct disclosure |
| Blibli third-party retail | 2022 | 37,050 | 199.00 | 36,851 | 0.54 | 185.2 | Direct disclosure |
| Blibli third-party retail | 2023 | 49,912 | 1,125 | 48,787 | 2.25 | 43.4 | Direct disclosure |
| Blibli third-party retail | 2024 | 54,210 | 1,384 | 52,826 | 2.55 | 38.2 | Direct disclosure |
| Blibli third-party retail | 2025 | 53,187 | 1,551 | 51,636 | 2.92 | 33.3 | Direct disclosure |
| Bukalapak Group | 2020 | 85,081,843 | 1,351,664 | 83,730,179 | 1.59 | 61.9 | Direct disclosure |
| Bukalapak Group | 2021 | 122,618,794 | 1,869,122 | 120,749,672 | 1.52 | 64.6 | Direct disclosure |
| Bukalapak Group | 2022 | 153,738,958 | 3,618,366 | 150,120,592 | 2.35 | 41.5 | Direct disclosure |
| Bukalapak Group | 2023 | 164,380,065 | 4,438,269 | 159,941,796 | 2.70 | 36.0 | Direct disclosure |

## Appendix B. The checks on the test of H1

### B.1 The checks

The main test is in Table 5.3. This appendix lists each check with both p-values where both exist. Sources: `h1_extended.py` (`tables/h1_extended_results.json`), `tables/test_battery.csv`, `tables/h1_quarterly_results.json`, `tables/h1_points_vs_log.json`, `tables/h1_thesis_drops.json`. The two rows

on how platforms book shopper discounts come from `research/option_space_2026-10-01/materials_2026-10-15/incentive_accounting_check.py` (classification and filing quotes in `INCENTIVE_ACCOUNTING.md`); their shuffle uses a different seed.

| Check | Indonesian and regional (median) | Foreign (median) | Mann–Whitney p | Shuffle p |
|---|---|---|---|---|
| Main test, 5 against 27 | 0.201 | 0.062 | 0.0038 | 0.0221 |
| 5 against the 8 chosen first | 0.201 | 0.055 | 0.0225 | 0.0497 |
| 5 against the 19 added | 0.201 | 0.070 | 0.0046 | 0.0175 |
| Without the 8 foreign platforms that book shopper discounts as marketing costs (exploratory, 15 Oct 2026) | 0.201 | 0.062 | 0.0035 | 0.0066 |
| Only the 11 foreign platforms that deduct shopper discounts from revenue (exploratory, 15 Oct 2026) | 0.201 | 0.075 | 0.0043 | 0.0052 |
| Growth divergence in points, 5 against 26 | 27.0 | 6.5 | 0.0055 | |
| Without Ozon, 26 foreign platforms | 0.201 | 0.060 | 0.0030 | 0.0215 |
| Without Zalando, 26 foreign platforms | 0.201 | 0.064 | 0.0045 | 0.0238 |
| Without Ozon and Zalando, 25 foreign platforms | 0.201 | 0.062 | 0.0036 | 0.0230 |
| Without Blibli third-party retail | 0.169 | | 0.0137 | 0.0364 |
| Without Bukalapak Group | 0.225 | | 0.0049 | 0.0070 |
| Without GoTo on-demand | 0.225 | | 0.0030 | 0.0071 |
| Without Grab on-demand | 0.194 | | 0.0094 | 0.0166 |
| Without Tokopedia | 0.169 | | 0.0164 | 0.0409 |
| Three marketplaces against the 8 chosen first, comparing pairs of years, absolute growth divergence in points | 38.917 | 7.105 | 0.0022 | 0.0606 |
| Three marketplaces against the 8 chosen first, one median per platform, absolute growth divergence in points (claims ledger C47, `build_numbers.py`) | | | 0.0242 | |
| Three marketplaces against the 8 chosen first, comparing pairs of years, absolute growth divergence in points (without Sea) | 38.917 | 4.192 | <0.0001 | 0.0083 |
| Three marketplaces against the 8 chosen first, comparing pairs of years, absolute change in log take rate | 0.252 | 0.055 | 0.0037 | 0.0303 |
| Three marketplaces against the 8 chosen first, comparing pairs of years, absolute change in log take rate (without Sea) | 0.252 | 0.033 | 0.0001 | 0.0167 |
| Three marketplaces, dropping all three 2022–23 comparisons (5 pairs), absolute growth divergence in points | 14.411 | 7.105 | 0.0248 | 0.2222 |
| Three marketplaces, dropping the largest comparison (7 pairs), absolute growth divergence in points | 15.738 | 7.105 | 0.0061 | 0.0606 |
| Three marketplaces against all 40 foreign pairs, absolute growth divergence in points | 38.917 | 7.373 | 0.0014 | 0.0121 |
| Five Indonesian and regional series including constructed Grab and Shopee (12 pairs) against the 8 chosen first, absolute growth divergence in points | 42.017 | 7.105 | 0.0006 | 0.0225 |
| Tokopedia and GoTo on-demand as one company, 4 against 27 | | | 0.003 | 0.007 |

| Check | Indonesian and regional (median) | Foreign (median) | Mann–Whitney p | Shuffle p |
|---|---|---|---|---|
| Quarterly pairs, 3 against 8 (same quarter a year earlier) | 0.071 | 0.049 | 0.19 | 0.33 |

Growth in revenue and growth in transaction value had a correlation of 0.10 in the Indonesian main sample and 0.98 in the foreign sample excluding Sea (`tables/claims_ledger.csv`, C48 and C49).

The quarterly sample is Grab on-demand, GoTo on-demand and Blibli against Sea, Airbnb, Booking, DoorDash, eBay, Etsy, Lyft and Uber. Airbnb and Booking use trailing four-quarter sums because they book revenue at check-in. DoorDash pairs ending in 2025Q4 or later are dropped because of an acquisition.

In the pairs of the Indonesian and regional group, the take rate stayed within about 10 percent (0.10 in log terms) in one third of pairs, against 67 to 69 percent in the foreign groups. The version of the test with growth divergence in points has 26 foreign platforms because Ozon has no pair in which both years pass the own-goods rule. The main test keeps Ozon's pair, whose later year passes, and the first rows above show that dropping it, or Zalando, leaves the result in place.

The comparison in points of the take rate (exploratory, `h1_points_vs_log.py`): median yearly change 1.01 points in the Indonesian and regional group and 0.98 points in the foreign group (26 platforms), Mann–Whitney p = 0.65; median share kept 1.59 percent against 16.93 percent.

### B.2 Where the 2023 break sits (exploratory)

At the top of the size distribution, sellers with revenue above Rp50 billion a year (about US$3.4 million at the 2022 rate) were 0.5 percent of BPS's marketplace sellers in 2022 and held between 18 and 39 percent of the marketplace slice's value, resting on 19 sample rows (`tables/part5_checks.json`).

Across the 34 provinces, count growth had a rank correlation of 0.10 (p = 0.59) with the share of new sellers and 0.35 (p = 0.04) with growth in sample size. In 23 provinces count growth exceeded what entry can explain, and these provinces carry 98.5 percent of the increase. The standard error of the national count implied by BPS's published relative standard errors is about 1.4 percent. The 306 thousand newly counted sellers are about 4.6 times the standard error of the change, and 2.3 times if the true error were twice as large (`tables/bps_frame_check.json`, `tables/rq_closing_checks.json`).

## Appendix C. Registered tests and their outcomes

A registered test was written down, dated and stored in the repository before the data it concerns were opened. The dates are those of the commits in `research/option_space_2026-10-01/PREREGISTRATION.md`. Exact wording and thresholds are in that file.

| Test | What it predicted | Registered | Outcome |
|---|---|---|---|
| E1 | Sellers without financial statements hold a smaller share of online value than of the count | 2 Oct 2026 | Holds, as a range (25.7 to 40.4 percent of value against 79.3 percent of the count) |
| E2 | The marketplace share of online value in 2022 is above the published 2023 figure | 2 Oct 2026 | Holds, as a range (22.6 to 25.6 percent) |
| E2b, E2c | Marketplace use did not rise between 2020 and 2024 | 2 Oct and 3 Oct 2026 | Hold (21.6, 19.8, 17.8 and 17.2 percent) |

| Test | What it predicted | Registered | Outcome |
|---|---|---|---|
| E3 | Shopee reaches at least as many sellers as Tokopedia | 2 Oct 2026 | Holds (7.5 against 2.2 percent of all sellers) |
| E4 | Entrants use chat or social media at least as often as earlier sellers | 2 Oct 2026 | Holds, with no power (nearly everyone uses them) |
| E5 | At least 21.5 percent of 2023 sellers began in 2023 (the entry test) | 2 Oct 2026 | Falsified (13.45 percent). Disclosure: BPS's 2024 publication, held at registration, has a table of 2024 sellers by start year; if the seller population were stable from 2023 to 2024, it would cap the 2023 share near 18 percent. The table was first used on 8 Oct, after the test ran on 7 Oct |
| E6 | The median change in sales of sellers already online is at most 10.36 percent | 2 Oct 2026 | Holds (median 0 percent) |
| E7 | The share of value held by sellers without statements fell by at most 5 points from 2020 to 2022 | 3 Oct 2026 | Falsified, narrowly (5.8 points; 3.3 on the second version) |
| E8 | Sellers with revenue of Rp300 million a year or more hold over half of marketplace value | 3 Oct 2026 | Holds (92.2 percent, middle scenario, an upper bound) |
| RBI | The marketplace part sold to final consumers is below Bank Indonesia's 2022 figure | 3 Oct 2026 | Holds in the low (0.16) and middle (0.68) scenarios; fails in the high one (1.62) |
| W1, W2 | Unregistered firms using social media keep written records and take digital payments more often | 3 Oct 2026 | Hold (29.9 against 12.3 percent; 11.5 against 2.7 percent) |
| W3 | Their mean annual sales are below Rp100 million (about US$6,600) | 3 Oct 2026 | Falsified (mean Rp111 million; median Rp60 million) |
| F1, F2 | Formal firms with a website take more e-payments and file taxes online more often | 3 Oct 2026 | Hold |
| F3 | More than one third of formal firms have neither a website nor online tax filing | 3 Oct 2026 | Holds (39.9 percent), after a disclosed coding fix: the first run read "yes, partially" as no for online filing and gave 13.0 percent; the questionnaire counts it as yes |
| X3 | Indonesia's share of firms with neither is the highest of the countries compared | 3 Oct 2026 | Withdrawn as a test (addendum of 16 Oct 2026): published e-filing shares seen at registration already capped every other country below Indonesia. Reported as descriptive (Malaysia 9.1, Thailand 8.7) |

| Test | What it predicted | Registered | Outcome |
|---|---|---|---|
| X2 | Firms with a website file taxes online more often, in at least 3 of 4 countries | 3 Oct 2026 | Holds in 4 of 4 |
| Q1 | The test of H1 with quarterly pairs rejects H1 | 7 Oct 2026, before it was run, but after the quarterly series had been built and the annual result was known, so it is a robustness check | Falsified (p = 0.19) |
| M1 | Markets value growth from more transactions more than growth from a higher take rate | 8 Oct 2026, before the market values were pulled | Falsified in the main sample (p = 0.20) and with extreme values capped (p = 0.16); holds in the second run of 34 platforms (p = 0.004), which mixes segment figures with whole-company values |
| M2 | Market value moves with transaction value | 8 Oct 2026 | Holds (p = 0.002) |
| L (P1 to P3) | Large take-rate moves driven by discounts occur only where incentives are at least as large as revenue (L of 1 or more) | 2 Oct 2026 | Thin support out of sample: P1 holds on 2 of 2 cases; P3 fails at the main threshold (median move 0.117 against 0.10); rank correlation 0.20 (p = 0.43) |
| P1 to P5, R1 to R3 | Predictions on the third-quarter 2026 results of GoTo, Grab and Sea | 2 and 3 Oct 2026 | Not yet scored; first results are due on 27 Oct 2026 |
| Marketplace tax start | The tax start slips again | 2 Oct 2026 | Withdrawn: collection began on 1 Oct 2026, before the prediction was registered (addendum of 15 Oct 2026) |
| BPS marketplace share | The next published marketplace share is at most 15.79 percent | 2 Oct 2026 | Not yet scored |

## Appendix D. The values behind Figures 5.3 to 5.5

**Growth in 2023, percent (Figure 5.3).** Household spending +9.4; Tokopedia transaction value −8.9; GoTo on-demand transaction value −10.5 (rests on an estimated 2022 value); Bank Indonesia e-commerce −4.7; Momentum Works marketplace estimate +3.5 (calculated from its published levels; its 2024 release states 3.7); Bukalapak transaction value +6.9; Blibli third-party retail transaction value +34.7; Tokopedia revenue +53.2; BPS e-commerce value +40.6; BPS online sellers +27.4. Sources: `tables/consistency_grid_growth.csv`, `tables/yardsticks_bi_bps.csv`, `tables/wedge_over_time.csv`, `tables/h1_groundwork.json`, company filings.

**Who the extra sellers were, percent (Figure 5.4).** Share of the 2022 count and share of the extra count: business opened before 2010, 18.3 and 40.5; owner aged 50 or over, 28.9 and 41.2; outside Java, 23.6 and 36.3; food and lodging or making goods, 38.5 and 53.0. Source: `tables/bps_newly_counted_profile.json`.

**Marketplace use and financial statements, percent of sellers (Figure 5.5).** Marketplace use: 21.6 (2020), 19.8 (2022), 17.8 (2023), 17.2 (2024, published). Complete financial statements: 23.5 (2020), 20.7 (2022), 15.2 (2023). Source: `tables/bps_descriptives.json`.

The figures are drawn from the tables named above (`papers/thesis_2026-10/figures/make_figures.py`). In each figure, a legend or label names every colour.

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